The Canine Rosetta Stone

How Far Are We from Translating What Dogs Say?

A Survey of Cross-Species Communication Decoding
in the Age of Foundation Models

The `woof-to-word` Project

An open, evidence-bounded literature survey

`github.com/mcxiaoxiao/woof2word`

November 2026 • Version 1.0

Abstract

Can a machine turn a bark into a sentence? Consumer products already claim to "translate" dog vocalizations with accuracies above 80%, while landmark projects such as Project CETI, Google's DolphinGemma, and the Earth Species Project apply foundation models to sperm whales, dolphins, and songbirds. This survey asks a deliberately narrow question: *how far are we, really, from translating what a dog says?* We organize the answer around a five-rung "translation ladder"—signal detection, affect classification, intent and context inference, referential semantics, and open-ended dialogue—and argue that today's systems sit firmly on the second rung, with scattered footholds on the third. We synthesize three bodies of evidence: (i) the behavioral biology of canine communication (barks, growls, DogFACS facial actions, body posture, and soundboard button use); (ii) the technical pipeline that would be required for genuine translation (self-supervised acoustic encoding, sequence-structure discovery, semantic grounding, and interactive validation); and (iii) the commercial landscape and its evaluation crisis. We identify a central *semantic gap*: dogs possess a rich affective and referential signaling system but lack the recursive, generative syntax that human translation presupposes, so "translation" is partly a category error. We further show that the field's headline accuracies are largely *unfalsifiable*, because no ground-truth "intended message" exists to score against. We close with a capability roadmap, an honest distance estimate, and an ethics framework for responsible interspecies AI. Our conclusion is neither dismissive nor utopian: useful, validated *affect-and-intent* translation is plausible within this decade, while faithful open-ended translation of canine "language" may remain permanently out of reach—not for lack of model scale, but for lack of a language to translate.

Keywords: animal communication, bioacoustics, foundation models, canine cognition, DogFACS, semantic grounding, interspecies interaction, evaluation, research ethics.

A note on scope and method. This is a narrative, evidence-bounded survey, not a PRISMA systematic review. Sources were gathered through structured web and literature search in November 2026; peer-reviewed publications, preprints, project reports, and journalism are distinguished throughout, and commercial accuracy claims are reported *as claims*, not as verified results. Section 12 treats the verification problem explicitly.

Contents

1 Introduction: The Oldest Silence **5**
1.1 A billion companions, one-sided conversations . . . 5
1.2 Why now: four converging currents . . . 6
1.3 Not only "how far" but "whether": alchemy, flight, or perpetual motion? . . . 6
1.4 The thesis in one paragraph . . . 6
1.5 Contributions and organization . . . 7

2 The Human Translation Stack: A Transformer Genealogy **7**
2.1 From symbolic phrase tables to neural sequence models . . . 7
2.2 The Transformer and the pretraining paradigm . . . 8
2.3 Multilinguality, low-resource transfer, and mined parallelism . . . 8
2.4 Speech-to-speech and unified multimodal translation . . . 8
2.5 Evaluation: the part animals do not have . . . 8
2.6 Scale, instruction-tuning, and multimodal LLMs . . . 8

3 Speech and Audio Foundation Models: The Acoustic Toolkit **9**
3.1 Self-supervised speech representation learning . . . 10
3.2 Recognition architectures and the alignment problem . . . 10
3.3 Corpora and benchmarks that made scale possible . . . 10
3.4 Neural audio codecs and discrete acoustic tokens . . . 11
3.5 The visual backbone analogy . . . 11
3.6 Bioacoustic foundation models . . . 11

4 Transfer Feasibility: What Survives Contact with Animal Signals **11**
4.1 Transfer learning: the formal picture . . . 12
4.2 Representation-learning foundations . . . 13
4.3 The symbol-grounding problem . . . 13
4.4 Compositionality, systematicity, and what probing reveals . . . 13
4.5 Generative and interactive methods: the validation loop . . . 13

5 Seven Lessons from Human Translation for Animal Decoding **14**
5.1 Lesson 1: Pretrain on unlabeled signal before anything else . . . 14
5.2 Lesson 2: Discretize before you translate . . . 14
5.3 Lesson 3: Bootstrap when no parallel corpus exists . . . 14
5.4 Lesson 4: Share structure across related systems . . . 15
5.5 Lesson 5: Evaluation precedes the claim . . . 15
5.6 Lesson 6: Grounding is the wall, not scale . . . 15
5.7 Lesson 7: Separate the interface from the measurement . . . 15

6 Is It Possible at All? Alchemy, Flight, or Perpetual Motion **16**
6.1 Existence proofs I: referential signals in the animal kingdom . . . 16
6.2 Existence proofs II: the dog is a willing, readable partner . . . 16
6.3 Why the upper rung may be alchemy: the language question . . . 16
6.4 The layered verdict . . . 17

7 What Dogs Actually Signal: The Biological Substrate 17
7.1 Vocalizations: barks, growls, whines . . . 18
7.2 The face: DogFACS and the "puppy dog eyes" effect . . . 18
7.3 The body: posture, tail, and gaze . . . 18
7.4 Reception: dogs understand more than they say . . . 19
7.5 Soundboards: the closest thing to canine "speech" . . . 19
7.6 The takeaway . . . 19

8 The Translation Ladder: What "Translating a Dog" Could Mean 20
8.1 Rung 1: signal detection and segmentation . . . 20
8.2 Rung 2: affect and arousal classification . . . 20
8.3 Rung 3: intent and context inference . . . 21
8.4 Rung 4: referential semantics . . . 21
8.5 Rung 5: open-ended dialogue . . . 22
8.6 Why the ladder matters . . . 22

9 The Technical Pipeline: From Waveform to (Something Like) Meaning 22
9.1 Stage 1: self-supervised acoustic encoding . . . 22
9.2 Stage 2: sequence-structure discovery . . . 23
9.3 Stage 3: semantic grounding . . . 23
9.4 Stage 4: interactive validation (playback) . . . 23
9.5 Maturity summary . . . 24

10 Lessons from Other Species: Whales, Dolphins, and Finches 24
10.1 Project CETI: structure is not meaning . . . 24
10.2 DolphinGemma and CHAT: grounding through interaction . . . 25
10.3 The Earth Species Project: from encoding to interaction . . . 25
10.4 What transfers to dogs—and what does not . . . 25

11 The Commercial Landscape: Translators That Are Not Translators 26
11.1 A brief history: from Bow-Lingual to PetGPT . . . 27
11.2 The wider field . . . 27
11.3 The audit: what the accuracy numbers actually measure . . . 27
11.4 Why the gap persists . . . 27

12 The Evaluation Crisis: Why the Numbers Cannot Be Trusted 28
12.1 The ground-truth problem . . . 28
12.2 What the vendor numbers actually are . . . 29
12.3 Unfalsifiability and the absent complaint . . . 29
12.4 Clever Hans, anthropomorphism, and the observer effect . . . 29
12.5 What a valid evaluation would require . . . 29
12.6 The limitations, consolidated . . . 30
12.7 A falsifiability criterion for the field . . . 30

13 So How Far Are We? An Honest Distance Estimate 30
13.1 Rung 1–2: essentially here, and useful . . . 31
13.2 Rung 3: a foothold, a decade of work . . . 31

13.3 Rung 4–5: possibly never, and that is informative . . . 31
13.4 Why "more scale" is not the answer . . . 31
13.5 The reframe: from translation to interpretation . . . 31

14 Ethics and Governance: Responsible Interspecies AI 32
14.1 Welfare first: do no harm . . . 32
14.2 Honesty: labeling interpretation as interpretation . . . 32
14.3 Privacy: the home as sensor . . . 33
14.4 The animal as partner, not oracle . . . 33
14.5 Situated in AI ethics and animal sentience . . . 33
14.6 A governance checklist . . . 33

15 A Research Agenda: What Would Actually Move the Needle 34
15.1 A public, grounded canine corpus . . . 34
15.2 Calibrated, multimodal affect models . . . 34
15.3 Closed-loop intent validation . . . 34
15.4 Structure discovery, with humility . . . 34
15.5 Engineered dialogue, not extracted dialogue . . . 35
15.6 What would falsify our thesis . . . 35

16 Conclusion 35
16.1 Limitations of this survey . . . 36

References 36

1 Introduction: The Oldest Silence

For at least fifteen thousand years—and by some accounts more than thirty thousand—humans have lived alongside dogs [191], and for nearly as long we have talked *at* them without ever being able to talk *with* them. The wish to close that loop is older than science: folktales grant the hero a magic fern-seed or a drop of dragon's blood that lets him understand the speech of beasts. What is new is the scale of the longing, and the arrival of a technology that claims to satisfy it.

1.1 A billion companions, one-sided conversations

The demographic facts make the silence hard to ignore. There are roughly **900 million dogs** on Earth [217], of which on the order of **half a billion** live as companions in human homes [218]. Dogs are the most widely kept companion animal on the planet: some **65 million U.S. households** own at least one [220, 217], about **a quarter of European households** keep a dog [219], and tens of millions more households do so across Asia [222]. And these households do not treat their dogs as mute appliances: in one national survey **58% of owners report regularly conversing** with their dog or cat, and the majority do so precisely because the animal is experienced as a family member [221]. Table 1 collects these figures. The picture they form is striking: *hundreds of millions of people hold daily, affectionate, one-sided conversations with a companion who cannot reply in any shared language.* That asymmetry—intimate cohabitation without mutual intelligibility—is the regret this survey takes as its starting point.

Table 1: The scale of the human–dog communication gap. Industry and survey estimates; orders of magnitude, not censuses.

Quantity	**Estimate**	**Source**
Dogs worldwide (all regimes)	≈900 million	[217]
Companion (pet) dogs, 2024	≈528 million	[218]
U.S. households owning ≥1 dog	≈65.1 million	[220]
European households owning ≥1 dog	≈25% (≈90M dogs)	[219]
Pet dogs, China	≈54 million	[217]
Owners regularly conversing with pet	58% (AU survey)	[221]

What is new is that the old wish now arrives packaged as a product. A smartphone application records a bark and returns a sentence—"I want to play," "my tummy hurts," "I love you"—in under a second. The vendor advertises an accuracy. The number is large. And the curious reader is left with two questions that turn out to be surprisingly hard to answer: *how far are we, actually, from translating what a dog says?* and, more fundamentally, *is the thing possible at all?*

This survey takes that question seriously and refuses to answer it with a single number. "Translation" is not one capability but a stack of them, and the distance to each layer is different. A system that reports whether a dog is relaxed or distressed is doing something real and useful; a system that claims to render the dog's "words" as an English sentence is doing something else, and the difference between the two is not a matter of degree but of kind. Our central move is to make that stack explicit—a five-rung *translation ladder* (Section 8)—and to locate current research and current products on it. The headline finding is easy to state and, we will argue, well supported: **today's best systems live on the second rung, affect**

classification, with a handful of careful laboratory studies reaching toward the third. Everything above that remains either aspirational or, in the strong sense, ill-posed.

1.2 Why now: four converging currents

Four developments make this the right moment for such a survey. First, *the human translation stack has matured* into a reusable toolkit: sequence-to-sequence learning [39, 40], attention [41], the Transformer [44], large-scale pretraining [79, 82], multilingual translation [62, 63], and speech/audio foundation models [114, 109, 137] (Sections 2–3). Second, *bioacoustic foundation models* have matured in their own right: self-supervised encoders trained on animal sound—AVES [24], the audio–language model NatureLM-audio [25], and Google DeepMind's Perch line [26]—now transfer across species and tasks, with standardized benchmarks such as BEANS [27]. Third, *flagship decoding projects* have produced concrete, peer-reviewed structure discoveries: Project CETI's sperm-whale "phonetic alphabet" [19], Google's DolphinGemma sequence model [21], and the Earth Species Project's interactive songbird work [29]. Fourth, *canine cognition* has its own empirical renaissance, from the ELTE Budapest group's decades of bark and growl psychophysics [1, 4] to the UC San Diego soundboard studies that put "talking buttons" under controlled observation [15, 16]. The intersection of these four currents is exactly the space this paper maps.

1.3 Not only "how far" but "whether": alchemy, flight, or perpetual motion?

A survey that only estimates distances implicitly assumes the destination exists. We refuse that assumption. History offers three templates for a long-cherished impossible-sounding goal. *Alchemy* pursued a target (transmutation of lead into gold by chemical means) that rested on a category error about the underlying science; no amount of effort or scale would reach it. *Heavier-than-air flight* was widely declared impossible by competent authorities, yet rested on sound physics and was achieved once the right mechanisms were assembled. *Perpetual motion* is provably impossible, ruled out by conservation laws. The central question we pose is which template interspecies translation fits (Section 6). Our answer is layered: the *lower rungs* (affect, intent) resemble flight—hard but grounded, with existence proofs already in hand (Section 6, Table 6); the *upper rung* (open-ended sentence-level translation of a non-linguistic signal) resembles alchemy—not because the engineering is hard but because the target may not exist. Distinguishing these two is the survey's most consequential act.

1.4 The thesis in one paragraph

A dog's communicative repertoire is rich, multimodal, and genuinely informative—but it is, in the technical sense, a *signaling system* rather than a *language*. It encodes affect, arousal, motivational state, individual identity, and some referential content (a specific toy, a specific door), yet it lacks the open-ended, recursive, generative syntax that human language translation presupposes [183, 185]. Consequently, "translating what a dog says" is, above the level of affect and intent, partly a *category error*: there may be no sentence-shaped meaning in the signal to recover. This is not a counsel of despair. It reframes the achievable goal—validated, calibrated affect-and-intent inference—and exposes why the field's most quotable numbers ("81.5% accuracy," "94.6% accuracy") are, as presently defined, close to meaningless: there is no ground-truth intended message to score a translation against. We call this the *falsifiability problem*,

and it is the methodological heart of the survey (Section 12).

1.5 Contributions and organization

This paper is written from a computer-science and AI standpoint, with biology, linguistics, and ethics as supporting witnesses. It makes six contributions.

(1) **A CS/AI tech-stack synthesis**: the human translation genealogy from seq2seq to Transformers to speech/audio foundation models, and what each layer contributes to animal decoding (Sections 2–3, Tables 2–3).

(2) **A transfer-feasibility audit and a set of lessons**: what human machine translation presupposes, what of it survives contact with animal signals, and what human MT teaches animal decoding (Sections 4–5, Table 4).

(3) **A feasibility adjudication**: alchemy vs. flight vs. perpetual motion, with a table of existence proofs (Section 6, Table 6).

(4) **A conceptual ladder** separating signal, affect, intent, reference, and dialogue, used to adjudicate over-claiming (Section 8, Figure 1).

(5) **A critical audit** of commercial "pet translators" and the evaluation crisis, with a consolidated limitations table (Sections 11–12, Table 11).

(6) **An honest distance estimate and roadmap**, with confidence bands and an ethics framework (Sections 13–15, Figure 2).

Section 7 grounds the discussion in what dogs actually signal; Section 16 concludes. Throughout, we keep two registers separate: what is *measured* (peer-reviewed effect sizes, model metrics on defined tasks) and what is *claimed* (vendor accuracies, press-release superlatives). The gap between those registers is, in a real sense, the subject of this paper.

2 The Human Translation Stack: A Transformer Genealogy

Before asking what machines can make of a bark, we must be precise about what machines already make of human speech and text. The last decade of machine translation (MT) is not a single invention but a layered stack, and each layer supplies a tool that animal decoding either inherits or lacks. This section reads that stack from a computer-science standpoint; Section 4 then audits which layers survive transfer to non-human signals.

2.1 From symbolic phrase tables to neural sequence models

Statistical MT treated translation as noisy-channel decoding over phrase tables learned from parallel corpora such as Europarl [53] and OPUS [54], toolkitted by Moses [52]. The neural turn replaced hand-built phrases with end-to-end encoder–decoder learning [39, 40], and the decisive ingredient was *attention*: a soft, learned alignment between source and target positions [41, 42]. Convolutional variants competed briefly [43], and production systems (GNMT) demonstrated that deep recurrent attention models could be deployed at scale [45, 48]. Two data-side inventions proved as important as the architecture: subword segmentation (BPE), which lets a model compose rare words from frequent pieces [50], and back-translation, which manufactures parallel signal from monolingual text [51, 49]. Both are directly relevant to animals: BPE is a recipe for discretizing a continuous signal into reusable units, and back-translation is a recipe for learning when no parallel corpus exists.

2.2 The Transformer and the pretraining paradigm

The Transformer dispensed with recurrence entirely, relying on self-attention [44]. Its real legacy, however, is that it made *pretraining* scalable: masked language modeling (BERT) [79], causal language modeling (GPT) [80, 81, 82], and sequence-to-sequence denoising (T5, BART) [83, 84] turned massive unlabeled corpora into general-purpose representations. Ablations and refinements (RoBERTa, XLNet, ELECTRA, DeBERTa) [85, 86, 87, 88] established that the objective and data regime matter more than architectural cleverness. Contextual embeddings had already shown the value of pretraining over static vectors (word2vec, GloVe, ELMo, ULMFiT) [75, 76, 77, 78]. For bioacoustics the lesson is that a large *unlabeled* corpus of vocalizations, pretrained with a masked or contrastive objective, is the right starting point—exactly the strategy later adopted by wav2vec 2.0 and AVES (Section 3).

2.3 Multilinguality, low-resource transfer, and mined parallelism

Human MT's most encouraging result for animal decoding is that translation quality does not require a bespoke parallel corpus per language pair. Shared-encoder multilingual models enable zero-shot transfer [46, 47]; cross-lingual pretraining (XLM, XLM-R, mBART, multilingual BERT) [91, 89, 74, 90] shares structure across languages; and massive mined parallelism (WikiMatrix, CCAligned) plus English-centric or many-to-many training (M2M-100, NLLB) brought hundreds of low-resource languages into scope [67, 68, 64, 62, 63]. Even fully *unsupervised* MT is possible from monolingual corpora alone [65, 66]. This is the closest human analogue to the animal situation: a target "language" with no parallel corpus, where one must bootstrap structure from raw signal and transfer from related, better-resourced systems.

2.4 Speech-to-speech and unified multimodal translation

Translation moved from text-only to end-to-end speech: listen-and-translate proofs of concept [69], direct sequence-to-sequence speech translation [70], and direct speech-to-speech models [71], culminating in unified multilingual multimodal systems (SeamlessM4T, Seamless) [72, 73]. These architectures—acoustic encoder plus semantic decoder plus optional vocoder—are precisely the template a genuine animal translator would need, and they show the template is engineerable *when the source is human language*.

2.5 Evaluation: the part animals do not have

Human MT rests on a century of accumulated *reference translations*: BLEU [55], METEOR [56], and learned metrics such as COMET [57], validated against expert human judgment [58] and adjudicated annually at WMT [59, 60, 61]. Every one of these presupposes a ground-truth target sentence. Animal communication has no such reference, which is why importing MT metrics wholesale is a category error (Section 12).

2.6 Scale, instruction-tuning, and multimodal LLMs

Finally, scaling laws make returns predictable [101], compute-optimal training clarifies the data/scale trade-off [102], and the foundation-model paradigm frames both opportunity and risk [100]. Open and proprietary LLMs (LLaMA, Llama 2, GPT-4, Gemini, Claude, Gemma) [92, 93, 94, 95, 96, 97], instruction tuning [98], and chain-of-thought reasoning [99]

supply the semantic and interactive layer; vision-language models (CLIP, Flamingo, LLaVA, BLIP-2) [103, 104, 105, 106] show how to bind a non-linguistic modality to language—the exact operation a dog translator would attempt when grounding a bark to a context. Table 2 condenses the genealogy and, for each layer, states what it offers animal decoding.

Table 2: The human translation / Transformer genealogy and its transfer value to animal decoding. "Transfer value" is our assessment, not a vendor claim.

Year	Milestone	Key mechanism	Data regime	Transfer value to animals
2006	CTC [115]	alignment-free sequence labeling	weak labels	label variable-length vocalizations
2014	seq2seq [39, 40]	encoder–decoder	parallel	map signal sequence → symbol sequence
2015	attention [41, 42]	soft alignment	parallel	align bark segments to labels
2016	BPE [50]	subword composition	monolingual	discretize continuous signal into units
2016	back-translation [51, 49]	synthetic parallelism	monolingual	bootstrap without a parallel corpus
2017	Transformer [44]	self-attention	large parallel	the universal backbone
2017	multilingual NMT [46, 47]	shared encoder, zero-shot	many pairs	one encoder across species
2019	BERT / GPT-2 [79, 81]	masked / causal pre-training	massive unlabeled	foundation encoder from raw signal
2020	GPT-3 / T5 / BART [82, 83, 84]	scale + text-to-text	massive	few-shot task framing
2020	scaling laws [101]	predictable returns	compute	budget planning
2021	ViT / CLIP [202, 103]	cross-modal transformers	image–text	bind non-linguistic modality to language
2022	NLLB / M2M-100 [62, 64]	200-language mining	mined parallel	low-resource transfer recipe
2022	Chinchilla [102]	compute-optimal	tokens	data-vs-scale trade-off
2023	LLaMA / GPT-4 / Gemini [92, 94, 95]	open + multimodal LLMs	massive	semantic & interactive layer
2023	Whisper [114]	weak-supervision ASR	680k hours	robust acoustic encoder
2023	SeamlessM4T [72]	unified speech/text MT	multilingual	end-to-end speech translation template
2024	NLLB (Nature) [63]	200 languages deployed	mined	proof that low-resource scale works

3 Speech and Audio Foundation Models: The Acoustic Toolkit

Section 2 traced the translation stack whose *source* is already human language. A bark is not. To turn a continuous, noisy, variable-length vocalization into something a translator could consume, one needs the acoustic front-end and the audio-token machinery that the speech community built for itself. This section catalogs that toolkit from a computer-science standpoint—self-supervised encoders, recognition architectures, the corpora and benchmarks that made scale

possible, neural audio codecs, and audio–language models—and asks, for each, what an animal decoder inherits. The recurring lesson is that *the engineering of the front-end is largely solved; the missing piece is semantic, not acoustic.*

3.1 Self-supervised speech representation learning

The decisive idea is to learn a general acoustic representation from raw, unlabeled waveform, then fine-tune on a small labeled task. Contrastive Predictive Coding (CPC) showed that predicting future latent frames yields transferable features [107]; wav2vec extended this with a contrastive objective over a discrete vocabulary [108]; and wav2vec 2.0 combined a convolutional feature encoder, a quantized latent space, and a masked Transformer to reach strong recognition with only minutes of transcribed audio [109]. HuBERT replaced the contrastive target with masked prediction of offline-clustered units [110], WavLM added denoising and speaker-aware pretraining [111], and data2vec unified the masked-prediction objective across speech, vision, and text [112], with a faster successor in data2vec 2 [113]. Two extensions matter directly for animals: wav2vec-U learns to recognize speech *without any transcriptions* by adversarially aligning discrete units to a language model [126], and multilingual scaling (XLS-R, MMS, USM) pushes a single encoder across dozens to thousands of languages [127, 128, 129]. The animal analogue is exact: a large unlabeled corpus of barks, pretrained with a masked or contrastive objective, is the correct starting point, and the transcription-free recipe is the closest human precedent for a target with no parallel corpus.

3.2 Recognition architectures and the alignment problem

Before Transformers, end-to-end recognition was built on recurrent networks with two alignment regimes. Connectionist Temporal Classification (CTC) marginalizes over all alignments between input frames and output labels [115], and deep recurrent CTC systems scaled to end-to-end speech recognition [116, 117]; attention-based listen-attend-spell models instead learn a soft alignment [118]. The Conformer married convolution with self-attention to capture both local and global acoustic structure [119]. For animal signals the alignment problem is the same in form—a variable-length vocalization must be mapped to a variable-length label sequence—but harder in substance, because there is no canonical label sequence to align to (Section 12).

3.3 Corpora and benchmarks that made scale possible

None of the above would exist without shared data and evaluation. LibriSpeech provided a thousand hours of read audiobook speech [120]; Common Voice crowdsourced a massively multilingual corpus [121]; Multilingual LibriSpeech extended read speech across languages [122]; and the older TIMIT and Switchboard corpora anchored phonetic and conversational research [123, 124]. Benchmarks then standardized what "good" means: SUPERB probes a frozen representation across many speech tasks [125], while CoVoST and CoVoST 2 supply speech-translation references [130, 131]. The bioacoustic community is only now building the equivalent infrastructure (BEANS, BirdSet; Section 10), and the absence of a "LibriSpeech of barks" is a concrete, fixable bottleneck.

3.4 Neural audio codecs and discrete acoustic tokens

A second lineage turns waveform into *discrete tokens* that a language model can predict. WaveNet modeled raw audio autoregressively [132]; VQ-VAE introduced a learned discrete latent codebook [133]; Jukebox showed that hierarchical VQ-VAEs could generate coherent music [134]. SoundStream and EnCodec made neural audio compression practical with residual vector quantization [135, 136]. AudioLM then demonstrated that a language model over these tokens can generate long-range coherent audio without any text [137], and the recipe generalized to music (MusicLM, MusicGen) and general sound (AudioGen) [138, 139, 140]. VALL-E treated text-to-speech as neural codec language modeling [141]; UniAudio pushed toward a single foundation model for many audio tasks [142]; and AudioPaLM, SALMONN, and Qwen-Audio fused speech/audio encoders with large language models [143, 144, 145]. This is precisely the discretization step a "bark alphabet" would require: if vocalizations can be compressed into reusable tokens, sequence models can search for structure—the strategy Project CETI applied to whale codas [19].

3.5 The visual backbone analogy

The same representation-learning revolution happened in vision, and the analogy clarifies what transfers. AlexNet's ImageNet win established deep convolutional features as general-purpose [214, 215, 216]; residual networks made very deep models trainable [201]; and the Vision Transformer later showed the Transformer backbone works on pixels as well as tokens [202]. DogFACS-style facial analysis (Section 7) rides directly on this lineage. The lesson is that a modality becomes tractable once a large pretrained backbone exists—and that the backbone, not the task head, is where the value lives.

3.6 Bioacoustic foundation models

Finally, the toolkit has been pointed at animals. BirdNET demonstrated large-scale, consumer-grade bird identification from audio [166]; comprehensive reviews chart both the promise and the methodological pitfalls of computational bioacoustics [167, 168]. Self-supervised AVES [24], the audio–language NatureLM-audio model [25], Google's Perch embeddings [26], the BEANS benchmark [27], and open general-purpose audio encoders such as OpenBEATs [28] together show that the human speech recipe—pretrain on raw audio, transfer to many tasks—works across species. What none of them supplies is a *reference translation*: they classify, embed, and retrieve, but they do not, and cannot yet, render a bark as a sentence with measured fidelity. Table 3 condenses the toolkit and its transfer value.

4 Transfer Feasibility: What Survives Contact with Animal Signals

We now have two stacks in view: the human translation genealogy (Section 2) and the speech/audio toolkit (Section 3). The decisive question is not whether these tools are powerful—they are—but which of their *presuppositions* survive when the source is a dog rather than a human. This section audits the transfer component by component, grounding the audit in the machine-learning literature on transfer, representation, grounding, and compositionality.

Table 3: The speech/audio foundation-model toolkit and its transfer value to animal decoding. "Transfer value" is our assessment. SSL = self-supervised; ASR = automatic speech recognition; codec = discrete acoustic tokens.

Model / family	Type	Objective / data	Transfer value to animals
CPC / wav2vec / wav2vec 2.0 [107, 108, 109]	SSL	contrastive / masked, raw audio	foundation bark encoder from unlabeled audio
HuBERT / WavLM / data2vec [110, 111, 112, 113]	SSL	masked unit prediction	robust, denoised vocalization features
wav2vec-U [126]	SSL	transcription-free	learn without a parallel corpus
XLS-R / MMS / USM [127, 128, 129]	SSL	100s–1000s of languages	one encoder across species
CTC / LAS / Conformer [115, 118, 119]	ASR	alignment-free / attention	map variable-length signal to labels
Whisper [114]	ASR	680k h weak supervision	robust off-the-shelf acoustic encoder
LibriSpeech / Common Voice / SUPERB [120, 121, 125]	data	corpora + benchmarks	template for a "bark corpus"
SoundStream / EnCodec [135, 136]	codec	residual VQ	discretize vocalizations into tokens
AudioLM / VALL-E / UniAudio [137, 141, 142]	codec-LM	token language modeling	search for sequence structure
AudioPaLM / SALMONN / Qwen-Audio [143, 144, 145]	audio-LLM	audio + text fusion	bind acoustic signal to language
AVES / NatureLM / Perch / BEANS [24, 25, 26, 27]	bioacoustic	animal audio SSL	cross-species transfer, already realized

The result, summarized in Table 4, is a clear pattern: *the acoustic and representational layers transfer almost intact; the semantic and evaluative layers do not transfer at all.*

4.1 Transfer learning: the formal picture

Transfer learning formalizes when knowledge from a source domain helps in a target domain [146, 147]. Deep features are often transferable, with generality decreasing as one moves from early to late layers [148]. This is the optimistic half of the story: an encoder pretrained on human speech or general audio can be fine-tuned on barks, and cross-species bioacoustic models already demonstrate the gain [24, 26]. But transfer of *features* is not transfer of *semantics*. The source domain of human MT carries a target language with shared reference; the animal target domain carries no such thing. Feature transfer succeeds precisely where the task is classification; it cannot manufacture a reference translation that the source domain never had.

4.2 Representation-learning foundations

The machinery that makes transfer possible is the standard deep-learning stack: backpropagation [212], long short-term memory for sequences [213], the deep-learning synthesis of LeCun, Bengio, and Hinton [209, 210, 211], and the optimization and regularization practices (Adam, dropout) that make large models trainable [199, 200]. These are modality-agnostic: nothing in them assumes human language. A bark encoder built from the same components is therefore engineerable, and indeed AVES and NatureLM-audio are built from exactly them [24, 25]. The foundations transfer without remainder; the bottleneck is downstream.

4.3 The symbol-grounding problem

Here the transfer breaks. Human translation is grounded: source and target symbols alike refer to a shared world of objects, actions, and states [159]. A model trained only on text inherits this grounding indirectly through human usage, but the limits of that inheritance are well documented—text-only models lack the sensorimotor experience that gives words their meaning [160]. For a dog there is no shared symbolic world to ground into. The dog's signal is grounded in *its* affective and motivational states, not in a public lexicon; mapping it to an English sentence is not translation between two grounded languages but an act of interpretation across an unbridgeable referential gap. This is the technical content of the "semantic gap" thesis, and it is why grounding, not model scale, is the binding constraint.

4.4 Compositionality, systematicity, and what probing reveals

Human language is compositional: the meaning of a whole is a function of the meanings of its parts and their combination. Whether neural models learn this is contested [161], and diagnostic benchmarks (SCAN, systematic-generalization tests) show that standard architectures often fail to generalize compositionally even when they fit the training distribution [163, 162]. Probing studies further show that what a model "knows" is task- and probe-dependent, and that BERT-style representations encode syntax without necessarily encoding meaning [164, 165]. The implication for animals is twofold. First, even if a bark sequence had compositional structure (Section 6), our models might not recover it. Second, the absence of compositionality in the *signal*—the likely case for dogs—means there is no recursive structure for a sequence model to exploit, so the very property that makes human MT work is missing at the source.

4.5 Generative and interactive methods: the validation loop

Two further ML lineages matter not for decoding but for *validation*. Generative models—GANs [203], denoising diffusion models [204], and latent diffusion [205]—could synthesize candidate vocalizations for playback experiments, closing the loop between hypothesis and animal response. Reinforcement learning—from DQN [207] through PPO [206] to the self-play paradigm of AlphaGo [208]—supplies the framework for an interactive agent that learns which inferred intents predict the dog's behavior. These are the tools that could turn an unfalsifiable interpretation into a measured, closed-loop system (Section 12). They transfer well; what they require is an error signal, which the field currently lacks.

Table 4: Transfer-feasibility audit: what each layer of the human translation stack presupposes, and whether that presupposition holds for animal signals. Verdicts are our assessment.

Component	Human MT presupposes	Animal reality	Verdict
Acoustic encoder	raw waveform, any language	raw waveform, any species	transfers
Self-supervision	large unlabeled corpus	large unlabeled barks exist	transfers
Subword / codec units	discrete reusable units	units discoverable (cf. CETI)	partial
Sequence model	recursive, compositional source	likely non-compositional signal	limited
Parallel corpus	bilingual reference text	no reference exists	fails
Grounding	shared public lexicon	private affective states	fails
Evaluation metric	ground-truth target sentence	no intended message to score	fails
Closed-loop validation	observable error signal	buildable via playback / RL	buildable

5 Seven Lessons from Human Translation for Animal Decoding

The previous sections catalogued tools and audited their transfer. This section distills the exchange into seven concrete lessons—what the history of human machine translation, read as a computer-science narrative, teaches anyone attempting to decode an animal. Each lesson pairs a human-MT finding with a prescription for animal work. Table 5 collects them.

5.1 Lesson 1: Pretrain on unlabeled signal before anything else

The single most robust finding of the last decade is that a large unlabeled corpus, pretrained with a masked or contrastive objective, beats a clever architecture trained from scratch [79, 109, 85]. The animal prescription is immediate: invest first in a large, open, unlabeled corpus of dog vocalizations and a self-supervised encoder, exactly as AVES and NatureLM-audio have done for animals generally [24, 25]. Architecture is not the bottleneck; data regime is.

5.2 Lesson 2: Discretize before you translate

BPE showed that composing rare words from frequent subword units unlocks generalization [50]; neural codecs showed the same for audio [135, 136]; and Project CETI's "phonetic alphabet" showed it for whales [19]. The prescription: before claiming to translate, discover whether barks admit a small set of reusable, recombinable units. If they do not, the signal is not language-like and the translation framing should be dropped.

5.3 Lesson 3: Bootstrap when no parallel corpus exists

Human MT learned to manufacture its own supervision: back-translation [51, 49] and fully unsupervised translation [65, 66] removed the need for a bespoke parallel corpus. Animals have no parallel corpus and never will. The prescription is to build a synthetic loop: generate candidate vocalizations, play them back, and score the animal's response—the generative-plus-reinforcement recipe of Section 4. The error signal must come from the animal, not from a human annotator's intuition.

5.4 Lesson 4: Share structure across related systems

Multilingual models showed that one encoder can serve hundreds of languages, with low-resource pairs benefiting from high-resource relatives [46, 62, 64]. The prescription: train across species and across individuals, letting wolves, coyotes, and many dogs inform a shared canid encoder, rather than fitting a bespoke model per dog. Cross-species transfer is the animal analogue of low-resource MT, and it is already paying off [26, 127].

5.5 Lesson 5: Evaluation precedes the claim

Human MT earned its credibility through reference corpora and shared tasks—BLEU, WMT, learned metrics validated against expert judgment [55, 59, 57, 58]. The animal field has no reference and therefore no comparable discipline. The prescription is the demarcation of Section 12: define the task, disclose the protocol, and refuse the word "translator" until an independent error signal exists. A field that cannot measure cannot improve.

5.6 Lesson 6: Grounding is the wall, not scale

Scaling laws make returns predictable [101, 102], which tempts the inference that a big enough model will eventually translate a bark. But the symbol-grounding problem is not a data deficit; it is a referential mismatch [159, 160]. No amount of compute creates a shared lexicon where none exists. The prescription: spend effort on grounding and validation, not on parameter count, for the upper rungs. Scale helps Rungs 1–3; it cannot fix a category error at Rung 5.

5.7 Lesson 7: Separate the interface from the measurement

Modern systems render fluent sentences with a language model [82, 98], and fluency is easily mistaken for fidelity—the hallucination and faithfulness literature documents exactly this failure [157, 158]. The prescription: treat the rendered sentence as a user-interface layer, clearly labeled as interpretation, and never let its fluency count as evidence. The honest number lives upstream, at the classifier; the sentence is packaging.

Table 5: Seven lessons from human machine translation, each paired with a prescription for animal decoding.

#	Human-MT finding	Prescription for animals
1	pretraining beats architecture [79, 109]	build a large unlabeled bark corpus + SSL encoder first
2	subwords / codecs unlock generalization [50, 19]	discover reusable bark units before claiming translation
3	unsupervised / back-translation [65, 51]	bootstrap supervision via playback + RL, not annotators
4	multilingual sharing helps low-resource [62, 46]	one shared canid encoder across species and individuals
5	references + shared tasks = credibility [55, 59]	no error signal, no "translator"; disclose protocols
6	scaling laws predict returns [101]	scale helps Rungs 1–3; grounding, not scale, blocks Rung 5
7	fluency $\neq$ fidelity [157, 158]	label the sentence as interface; score the classifier

6 Is It Possible at All? Alchemy, Flight, or Perpetual Motion

A survey that estimates *how far* we are from a goal quietly assumes the goal exists. We refuse that assumption here. The wish to understand animals is old enough to have acquired the structure of a myth, and myths come in three kinds. *Alchemy* chased a target—lead into gold by chemical means—that rested on a category error; effort and scale could never reach it. *Heavier-than-air flight* was declared impossible by competent authorities, yet rested on sound physics and fell once the right mechanisms were assembled. *Perpetual motion* is provably impossible, excluded by conservation laws. The question this section adjudicates is which template interspecies translation fits. Our answer is layered, and it is the survey's most consequential claim: **the lower rungs are flight, the upper rung is alchemy-like, and nothing here is perpetual motion.**

6.1 Existence proofs I: referential signals in the animal kingdom

The strongest evidence that decoding animal signals is *flight* rather than alchemy is that it has already been done, repeatedly, in other species. Vervet monkeys produce predator-specific alarm calls that elicit appropriate escape responses, and habituation–dishabituation experiments show the calls function as referential labels [171, 172]. Prairie dogs encode predator type, size, and color in their alarm calls [177], and chicken alarm calls carry graded referential content [178]. The honeybee waggle dance communicates distance and direction to food [173]; bottlenose dolphins use signature whistles as individual labels [169, 170]; and chimpanzees inform conspecifics of danger in a goal-directed way [174]. Most strikingly, Japanese tits combine calls into compositional sequences [175], and formal-semantics analyses find context-dependent, rule-governed meaning in animal calls [176]. These are existence proofs: referential and compositional signaling *exists* in nature, so decoding it is an engineering problem, not a category error. The open question is whether *dogs* sit in this referential class—and the canine evidence below suggests they sit closer to the affective end.

6.2 Existence proofs II: the dog is a willing, readable partner

Domestication did not merely tame the dog; it reshaped its social cognition for life with humans [187, 191]. Dogs follow human pointing and gaze better than chimpanzees [189], look back to humans when faced with an unsolvable task [188], and show reward-system activation to praise in fMRI [190]. They possess voice-sensitive brain regions homologous to humans' [37], and their tail-wagging lateralizes by emotional valence [36]. A few dogs learn hundreds of object names [13, 14], and soundboard-trained dogs produce non-random, non-imitative two-button combinations and respond to human button presses above chance [15, 16, 18]. Together these prove that the dog is not a black box: it perceives human communicative intent, signals its own states readably, and in controlled settings exchanges referential tokens. Affect-and-intent decoding therefore has empirical ground to stand on—it is flight with a runway already built.

6.3 Why the upper rung may be alchemy: the language question

The pessimistic half of the verdict concerns Rung 5, open-ended sentence-level translation. The comparative evidence suggests that the recursive, generative syntax human translation presupposes—the "faculty of language in the narrow sense"—may be uniquely human [183, 185, 179]. The language instinct, on this view, is a species-typical adaptation [184] rooted in

uniquely human shared intentionality [180], and the cultural-evolution literature shows how iterated learning and cumulative culture shape linguistic structure into something no individual animal invents [181, 182]. Even the statistical universals of human language (e.g. Zipf's law) are signatures of a generative system [186]. If a dog's repertoire lacks this recursive structure, then "translating its sentences" is alchemy-like: not hard engineering but a target that may not exist. We stress the modality: this is *not* a perpetual-motion proof. No conservation law forbids richer animal signaling, and the compositionality results above keep the empirical question open. The honest statement is that Rung 5 is *possibly ill-posed*, not *provably impossible*.

6.4 The layered verdict

Pulling the threads together: Rungs 1–2 (signal, affect) are achieved flight; Rung 3 (intent) is flight with a foothold; Rung 4 (reference) is flight in some species and an open empirical question for dogs; Rung 5 (dialogue) is alchemy-like, possibly a category error. Crucially, none of this is perpetual motion—there is no theorem ruling out interspecies understanding, and the existence proofs guarantee that *something* real is achievable. The practical consequence is that effort should be poured into the grounded, falsifiable lower rungs, while the upper rung is treated as a hypothesis about the animal rather than a product roadmap. Table 6 lists the existence proofs and their template verdicts.

Table 6: Existence proofs and the feasibility template each supports. "Flight" = hard but grounded and (partly) achieved; "alchemy-like" = the target may not exist; nothing here is "perpetual motion" (provably impossible).

Claim	**Existence proof**	**Template**
Affect is decodable (Rung 2)	humans classify barks/growls [1, 4]; tail lateralization [36]; voice regions [37]	flight (achieved)
Intent is inferable (Rung 3)	pointing comprehension [189]; look-back [188]; soundboard combos [16]	flight (foothold)
Reference exists in animals (Rung 4)	vervet & prairie alarms [171, 177]; dolphin labels [169]; bee dance [173]	flight (species-dependent)
Compositionality exists (Rung 4)	tit compositional calls [175]; formal semantics [176]	flight (rare)
Dogs read human speech	fMRI reward [190]; word learning [13, 14]	flight (achieved)
Open-ended translation (Rung 5)	no recursive syntax in dogs; narrow faculty possibly unique [183, 185]	alchemy-like
Provably impossible?	no conservation law forbids signaling; question stays open	not perpetual motion

7 What Dogs Actually Signal: The Biological Substrate

Before asking whether a machine can translate a dog, we must ask what there is to translate. Canine communication is multimodal, graded, and heavily context-dependent. It is also, crucially, *asymmetric*: much of what a dog "says" is addressed to humans, and the dog's receptive understanding of human signals far exceeds its productive repertoire. This section reviews the channels and the evidence that each carries decodable information. Table 7 summarizes the landscape.

7.1 Vocalizations: barks, growls, whines

The domestic dog is unusually vocal for a canid, a trait widely interpreted as a product of domestication and paedomorphosis. The ELTE Budapest ethology group has built the most systematic body of evidence on what barks encode. Pongrácz and colleagues showed that human listeners can classify barks recorded in different behavioral situations above chance [1], and that acoustic parameters—fundamental frequency, tonality, and inter-bark interval—carry emotional information that listeners reliably extract [2]. Later work established that barks also carry *individual identity*: listeners can discriminate dogs from bark acoustics alone [3], and machine classifiers applied to thousands of barks from a kennel of Hungarian sheepdogs recovered both context and individual at rates well above chance, with "stranger at the fence" and "fight" contexts most separable and "play" least. Growls are even more informative: Faragó et al. demonstrated that human listeners judge a dog's size and aggressive intent from growl formants, and that growls produced in food-guarding versus play contexts are acoustically and perceptually distinct [4]. Whines and whimpers, associated with separation and submission, complete a repertoire that is graded rather than discrete.

The engineering literature has begun to automate exactly these judgments. The ICASSP 2018 study "What is my dog trying to tell me?" framed bark analysis as automatic recognition of *context* and *perceived emotion* [5], explicitly building on the ELTE psychophysics. The important nuance—which we will return to in Section 12—is that the target label is the *human-perceived* emotion or the *eliciting context*, not the dog's "intended message." The model learns a mapping that humans already agree on; it does not recover a hidden sentence.

7.2 The face: DogFACS and the "puppy dog eyes" effect

Canine facial behavior is codable with the same rigor as human facial action coding. DogFACS, the Dog Facial Action Coding System [6], adapts Ekman and Friesen's FACS to the musculature of the dog face, defining action units (e.g., AU101, the inner brow raiser) and ear action descriptors. Using DogFACS, Kaminski et al. showed that dogs produce markedly more facial movement—especially the inner brow raiser that enlarges the apparent eye—when a human is *attending* to them than when the human's back is turned, and that the presence of food does not drive the effect [7]. This is a striking result: it suggests the "puppy dog eyes" expression is a *communicative* signal tuned to human attention, possibly shaped by selection during domestication, rather than a mere reflex of internal state. A 2024 comparative study further reported that domestication *constrains* the dog's facial expressiveness relative to wolves [8], complicating any simple "dogs emote freely" narrative.

Automated facial analysis is now an active computer-vision subfield. Explainable CNN pipelines recognize emotional states from canine facial action patterns [9], improved optimization schemes push expression-recognition accuracy [10], and 2025 work automates emotional-expression analysis end to end [11] and standardizes facial landmark detection [12]. These are genuine advances—but, again, their labels are human-coded affective categories or DogFACS action units, not propositions.

7.3 The body: posture, tail, and gaze

A large fraction of everyday canine signaling is postural: ear position, tail carriage and wag direction, weight distribution, piloerection, and gaze. Classic work reported lateralized tail-

wagging biases tied to approach versus withdrawal motivation, and the broader ethogram of dominance, submission, play (the play bow), and stress is well documented in the clinical literature. Body signals are often *more* diagnostic than vocalizations, a point commercial vendors themselves concede: Traini's founder has stated that audio alone is insufficient and that combining body language and facial expression raises accuracy. This is an implicit admission that the "translation" is really multimodal affect classification. Olfactory signaling—urine marking, pheromonal communication—is enormously important to dogs but essentially inaccessible to current consumer sensors, and we flag it as a blind spot rather than a solved channel.

7.4 Reception: dogs understand more than they say

Any honest account must separate what dogs *produce* from what they *comprehend*. The receptive side is impressive. Rico, a border collie, provided the first evidence of "fast mapping"—inferring a novel word's referent by exclusion—in *Science* [13]. Chaser, another border collie, was documented retrieving 1,022 named objects and responding to simple noun–verb combinations [14]. These are exceptional, intensively trained individuals, but they establish an upper bound on referential comprehension that is far higher than the productive repertoire. The asymmetry matters: a future "dialogue" system may succeed by letting dogs *receive* structured human input (as soundboards do) long before it can *read* structured canine output.

7.5 Soundboards: the closest thing to canine "speech"

The most direct attempt to give dogs a productive symbolic channel is the augmentative interspecies communication (AIC) soundboard: a grid of buttons that each speak a recorded word when pressed. Popularized by speech pathologist Christina Hunger and the Instagram dog Bunny, the phenomenon is now under controlled study by Federico Rossano's UC San Diego *TheyCanTalk* project, which tracks thousands of button-using pets across dozens of countries. Two 2024 peer-reviewed papers anchor the science. Bastos et al. (*PLoS ONE*) showed that soundboard-trained dogs respond appropriately to words such as "play" and "outside" whether the word is spoken by the owner or triggered by an unfamiliar button-presser, indicating word comprehension rather than cue-reading [15]. A companion *Scientific Reports* paper analyzed two-button combinations and found them to be non-accidental, non-random, and non-imitative—evidence of intentional, if linear, combination [16]. A 2025 review synthesizes the evidence and the skepticism [17]. The consensus interpretation is careful: the combinations look like associative sequences (name + action, place + go), not recursive grammar; spatial and immediate concepts ("outside," "paw") are robust, while temporal abstractions ("yesterday," "tomorrow") are rare and inconsistent. Soundboards thus demonstrate *referential intent* on the third rung of our ladder—but they do so by giving the dog a human-designed symbol set, not by decoding the dog's natural signal.

7.6 The takeaway

The biological substrate is real and rich: dogs signal affect, arousal, identity, motivation, and—with human scaffolding—referential intent. What the substrate does *not* contain is an open-ended generative grammar. This single fact, more than any limitation of current models, bounds what "translation" can mean. We formalize that bound next.

Table 7: Canine communication channels, the information they carry, and the strongest evidence. "Decodable today" refers to validated, above-chance machine or human inference on a defined label—not to sentence-level translation.

Channel	Information encoded	Key evidence	Decodable today
Bark	Affect, arousal, context, individual identity	Pongrácz et al. [1, 2]; Molnár et al. [3]	Affect/context (above chance)
Growl	Body size, aggressive intent, resource-guarding vs. play	Faragó et al. [4]	Size/intent (human + ML)
Whine/whimper	Separation distress, submission	Ethological observation	Coarse affect
Face (DogFACS)	Attention-directed signals, affective state	Waller et al. [6]; Kaminski et al. [7]	Action units; affect (CNN)
Body/tail/gaze	Motivation, stress, play, social intent	Clinical ethograms; wag-laterality studies	Posture classes
Olfaction	Identity, reproductive state, territory	Scent-marking literature	*Not* sensor-accessible
Reception (input)	Human word referents (up to 1,022)	Kaminski et al. [13]; Pilley & Reid [14]	Comprehension, not production
Soundboard (AIC)	Referential intent via human symbols	Bastos et al. [15, 16]	Word-level intent (trained)

8 The Translation Ladder: What "Translating a Dog" Could Mean

The phrase "translate what a dog says" smuggles in an assumption: that there is a sentence-shaped meaning waiting in the signal, the way a French sentence waits behind its English translation. The biological review of Section 7 gives us reason to doubt this. To make progress we replace the single verb "translate" with a ladder of five distinct capabilities, each with its own input–output contract, its own notion of ground truth, and its own distance from the present (Figure 1).

8.1 Rung 1: signal detection and segmentation

The base rung asks only: *is there a vocalization, where does it begin and end, and who produced it?* This is a classical detection-and-classification problem with objective ground truth (a human annotator marks the waveform), and it is essentially solved. Bioacoustic detectors trained on large corpora [26, 24] segment and attribute calls reliably; individual identification from bark acoustics is above chance [3]. Nothing about Rung 1 requires understanding.

8.2 Rung 2: affect and arousal classification

The second rung maps a signal to a small set of affective categories—happy, fearful, aggressive, distressed, relaxed—or to continuous arousal/valence dimensions. This is where nearly all commercial "translators" actually live, and where the peer-reviewed ML literature is strongest, because the labels are human-perceived emotions or eliciting contexts that annotators can agree on [2, 5, 9]. Rung 2 is *falsifiable*: one can hold out recordings, score against consensus labels,

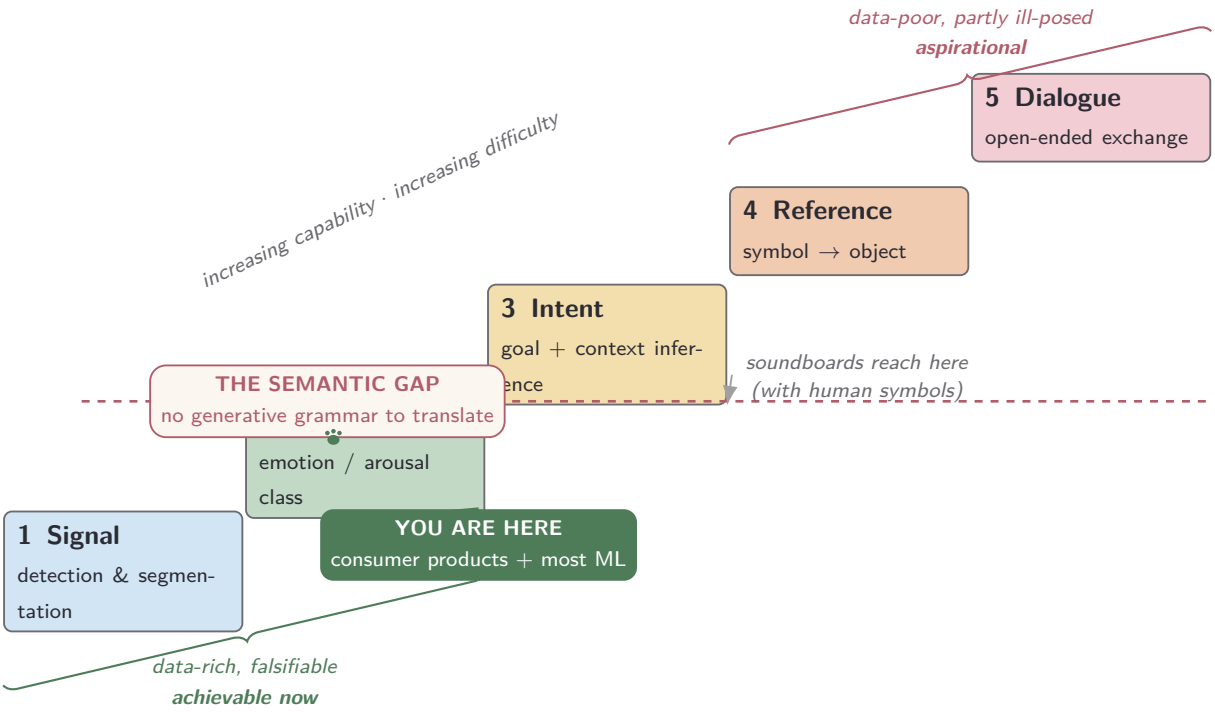


Figure 1: The translation ladder. "Translating a dog" decomposes into five capabilities of increasing difficulty. Current consumer products and most machine-learning work occupy **Rung 2** (affect classification); controlled soundboard studies reach **Rung 3** (referential intent) but only by supplying a human-designed symbol set. The dashed boundary marks the *semantic gap*: above it, the dog's natural signal lacks the recursive, generative syntax that sentence-level translation presupposes, so the target becomes partly ill-posed rather than merely hard. The lower rungs are data-rich and falsifiable; the upper rungs are data-poor and, in the strong sense, may have no ground truth to recover.

and report a real accuracy. It is also genuinely useful for welfare monitoring. Its limit is that an emotion label is not a sentence; rendering "fearful" as "I am scared of the vacuum" adds narrative the signal never contained.

8.3 Rung 3: intent and context inference

The third rung asks what the dog is *trying to do*—solicit play, request egress, guard a resource, signal pain—given the surrounding context. This requires multimodal fusion (voice + face + posture + situation) and begins to need a model of the dog's goals. Soundboard studies provide the cleanest evidence here, because the button press makes the intent explicit and checkable [15, 16]. But note the scaffolding: the symbol set is human-designed and trained. Rung 3 on *natural* signals is far less developed and much harder to validate.

8.4 Rung 4: referential semantics

The fourth rung would map a spontaneous canine signal to a specific referent—*that* ball, *this* door, a particular person—without human-supplied symbols. Receptive studies show dogs can *understand* such mappings [13, 14], and some vocal signals carry identity information, but a productive, open referential vocabulary in natural dog communication has not been demonstrated. This is the first rung where the "translation" framing starts to strain: there may be

no stable symbol-to-referent dictionary to recover.

8.5 Rung 5: open-ended dialogue

The top rung is the folkloric dream: free, generative, two-way conversation in which the dog expresses novel propositions and understands ours. It presupposes a recursive generative grammar on the canine side. The evidence of Section 7—graded affective signals, linear associative combinations, no syntax—indicates this presupposition is false for natural canine communication. Rung 5 is therefore not merely distant; in its strong form it is *ill-posed*. A weaker, engineered version (structured human-to-dog input via soundboards, with the dog responding through trained symbols) is conceivable and already nascent.

8.6 Why the ladder matters

The ladder converts a vague question into answerable ones. "Can we translate a dog?" becomes: *which rung, with what ground truth, at what validated accuracy?* It also explains the rhetoric of the market: vendors demonstrate Rung 2 while marketing Rung 5. The rest of this survey uses the ladder as a ruler—to measure the technical pipeline (Section 9), to read the cross-species evidence (Section 10), and to audit the products (Section 11).

9 The Technical Pipeline: From Waveform to (Something Like) Meaning

If genuine translation were possible, what would the system look like? Drawing on the cross-species decoding literature, we decompose the problem into four stages—*encoding*, *structure discovery*, *semantic grounding*, and *interactive validation*—and assess the maturity of each for the canine case. The stages are not a waterfall so much as a loop: validation feeds back into grounding, which constrains structure, which shapes the encoder.

9.1 Stage 1: self-supervised acoustic encoding

The modern starting point is a self-supervised representation trained on large, unlabeled audio. AVES (Animal Vocalization Encoder based on Self-Supervision) adapts the HuBERT/wav2vec paradigm to animal sound and shows that representations learned without labels transfer to downstream bioacoustic tasks [24]. NatureLM-audio extends this to an *audio–language* foundation model: trained on text–audio pairs spanning bioacoustics, speech, and music, it can detect vocalizations, classify rare species, and even caption or generate animal sounds from natural-language prompts, demonstrating transfer from speech and music to bioacoustics [25]. Google DeepMind's Perch line and the global birdsong embeddings show that a single encoder can serve many taxa [26]. Standardized benchmarks—BEANS (the Benchmark of Animal Sounds) and BirdSet—give these encoders a common evaluation surface [27], while open general-purpose encoders such as OpenBEATs push state of the art on a majority of BEANS subtasks [28]. For dogs, the implication is that a strong *Rung 1–2* encoder is available off the shelf; the bottleneck is not representation quality but labels and semantics.

9.2 Stage 2: sequence-structure discovery

Given an encoder, the next question is whether the signal has *combinatorial structure*—discrete units that recombine, the way phonemes form words. This is where the flagship projects have produced their most cited results. Project CETI's analysis of 8,719 sperm-whale codas revealed a "phonetic alphabet": rhythm, tempo, rubato, and ornamentation combine contextually, expanding the repertoire of distinguishable forms by nearly an order of magnitude [19]. Crucially, the authors are explicit that this describes *form*, not *meaning*: the paper answers what the units are and how they combine, not what most of them say. Google's DolphinGemma takes the language-model route directly—a roughly 400-million-parameter Gemma-based model with a SoundStream tokenizer, trained on the Wild Dolphin Project's four-decade Atlantic spotted dolphin corpus, that predicts the next sound in a sequence exactly as an LLM predicts the next token [21]. CETI's WhAM (2025) similarly learns to complete masked clicks and to render whale-like codas from other audio. The Earth Species Project's 2026 zebra-finch model (ZF-AIM) goes further, detecting, predicting, and *generating* calls in real time and engaging birds in turn-taking exchanges.

The lesson for dogs is double-edged. On one hand, the methods are species-agnostic and could be applied to bark sequences. On the other, the structure these models find is *statistical*, and predicting the next sound is not understanding it. Yossi Yovel and others have warned that such models may cluster calls by acoustic similarity while missing meaning entirely—grouping signals that sound alike but signify different things. For a repertoire as graded and context-bound as the dog's, the prior probability of finding a clean combinatorial alphabet is lower than for whale codas or birdsong.

9.3 Stage 3: semantic grounding

Structure without semantics is syntax without a dictionary. Grounding ties a signal to the world: to the eliciting context, the dog's subsequent behavior, the objects in view, the social partner. This is the hardest stage and the one where dogs are paradoxically *advantaged*, because they live in richly instrumented human homes and their behavior is legible to us. The soundboard paradigm is effectively a grounding device [15, 16]: the button press, the spoken word, and the consequent action (door opens, toy appears) form a triangulated, checkable mapping. The Wild Dolphin Project's CHAT system does the analogous thing underwater, pairing a synthetic whistle with a coveted object (sargassum, a scarf) so that the dolphin can request it by name [21]. The general principle—*ground symbols through interaction and consequence*, not through passive correlation—is the field's most important methodological insight, and it is exactly what consumer bark-to-sentence apps lack.

9.4 Stage 4: interactive validation (playback)

The final stage closes the loop: does the animal treat the model's output as meaningful? The gold standard is the *playback experiment*—play a recorded or synthesized signal and observe whether the receiver responds appropriately. Comparative ethology has used playbacks for decades to establish that, e.g., vervet alarm calls refer to specific predators. The Earth Species Project's interactive finch work and the CHAT dolphin protocol are modern, model-driven versions. For dogs, a validation loop would look like: infer an intent, emit a candidate signal or action, and test whether the dog's behavior confirms the inference—with the experimenter

blind to the model's guess to avoid Clever-Hans contamination. No consumer product does this; it is the dividing line between science and entertainment.

9.5 Maturity summary

Table 8 rates each stage for the canine case. The pattern is clear: the *front* of the pipeline (encoding, structure) is borrowable from the broader bioacoustics boom and is relatively mature; the *back* (grounding, validation) is where canine-specific, falsifiable work is needed and where almost none of the commercial effort is directed. This asymmetry—strong models, weak semantics, absent validation—is the technical signature of the whole field.

Table 8: Maturity of the four decoding stages for the canine case. "Borrowable" means general bioacoustic methods transfer directly; "canine-specific" means dedicated dog data and protocols are required.

Stage	Representative method	Maturity	Canine status
Encoding	AVES, NatureLM-audio, Perch [24, 25, 26]	High	Borrowable; Rung 1–2 ready
Structure	CETI alphabet, DolphinGemma, WhAM [19, 21]	Medium	Statistical only; meaning open
Grounding	Soundboards, CHAT object-labeling [15, 21]	Low–medium	Needs interaction data
Validation	Playback / closed-loop response	Low	Absent in products

10 Lessons from Other Species: Whales, Dolphins, and Finches

Dogs are not the first species for which decoding has been attempted, and the comparison is instructive. The flagship projects differ from the canine case in one decisive respect: their subjects often possess *discrete, culturally transmitted vocal units* (whale codas, dolphin signature whistles, birdsong syllables) that invite combinatorial analysis. Dogs, by contrast, signal with graded, context-bound affective displays. Reading the cross-species literature through the ladder of Section 8 clarifies both what is transferable and what is not. Figure 2 places these milestones on a capability timeline; Table 9 gives the details.

10.1 Project CETI: structure is not meaning

The Cetacean Translation Initiative is the field's most ambitious program. Its 2024 *Nature Communications* analysis of 8,719 codas from roughly 60 East Caribbean sperm whales demonstrated a context-dependent combinatorial system—rhythm, tempo, rubato, and ornamentation recombine to expand the form inventory by nearly an order of magnitude [19]. The 2022 program paper lays out the interdisciplinary ambition [20], and the 2025 WhAM model learns to complete and generate whale-like codas [22]; a late-2025 preprint even proposes a translative model of whale vocalization [23]. Yet the investigators are careful: the "phonetic alphabet" describes how signals are *built*, not what they *mean*. For dogs, the transferable lesson is the methodology (large corpora, structure discovery, playback validation); the *non*-transferable part is the assumption of a discrete combinatorial inventory, which canine graded displays may simply lack.

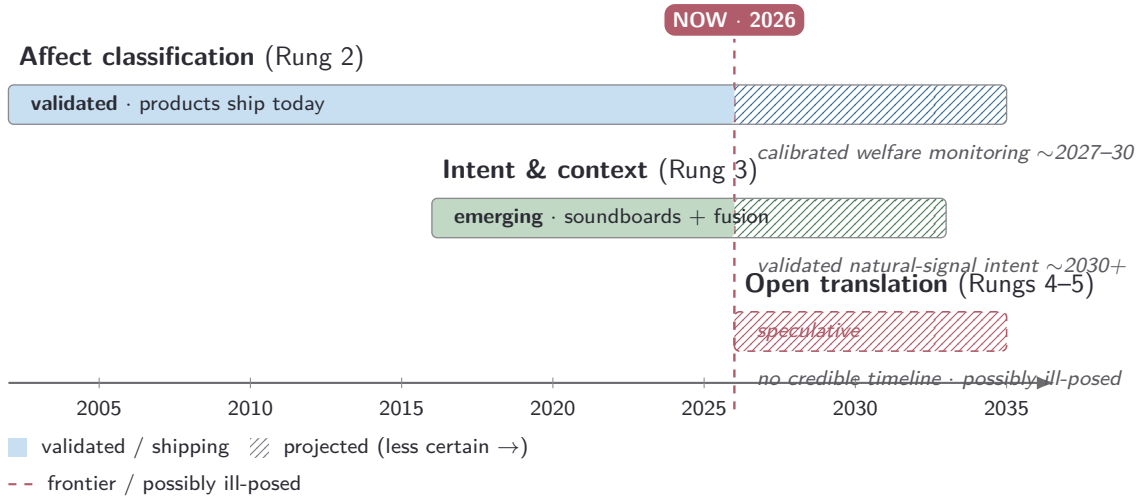


Figure 2: Capability roadmap, 2002–2035. Three tracks track the translation ladder over time. Solid bars denote validated, shipping capability; hatched bars denote projection, with confidence decreasing to the right. **Track A** (affect classification, Rung 2) is mature today and heads toward calibrated welfare monitoring. **Track B** (intent and context, Rung 3) is emerging through soundboards and multimodal fusion, with validated natural-signal intent plausible after 2030. **Track C** (open referential translation and dialogue, Rungs 4–5) has no credible timeline and may be permanently ill-posed for natural canine communication. The vertical dashed line marks the present (2026). Method-enabling milestones (CETI, DolphinGemma, foundation models) are detailed in Table 9; they advance the *tools*, not the *semantics*.

10.2 DolphinGemma and CHAT: grounding through interaction

Google DeepMind's DolphinGemma, with Georgia Tech and the Wild Dolphin Project, is a roughly 400-million-parameter audio sequence model built on Gemma with a SoundStream tokenizer, trained on four decades of Atlantic spotted dolphin recordings and small enough to run on a Pixel phone [21]. Its companion CHAT device does not try to translate natural whistles directly; instead it assigns *synthetic* whistles to coveted objects and lets dolphins request them by name—a grounding-through-interaction protocol. This is the single most important design lesson for canine work: meaning is established by *consequence* (the dolphin gets the sargassum), not by passive correlation. A dog soundboard is the terrestrial analogue.

10.3 The Earth Species Project: from encoding to interaction

The Earth Species Project (ESP), founded in 2018, supplies much of the field's open infrastructure: the AVES self-supervised encoder [24], contributions to NatureLM-audio and the BEANS benchmark [25, 27], and a 2026 zebra-finch model (ZF-AIM) that detects, predicts, and generates calls in real time, engaging birds in turn-taking exchanges [29]. ESP's trajectory—encode, then discover structure, then *interact*—is the pipeline of Section 9 made flesh. It also surfaces the standing critique: Yossi Yovel and others caution that models may over-rely on acoustic similarity, clustering calls that sound alike but mean different things [30].

10.4 What transfers to dogs—and what does not

Three things transfer: the self-supervised encoders, the structure-discovery toolkit, and above all the interaction-grounding/validation protocol. Three things do not: the assumption of

discrete culturally transmitted units, the luxury of decades of labeled wild corpora, and the clean playback accessibility of whales and birds. Dogs are domestic, multimodal, and graded; their advantage is legibility and proximity, their disadvantage is the absence of a syntax to decode. The cross-species evidence thus *supports* the ladder's verdict: the tools for Rungs 1–3 are arriving fast, while Rungs 4–5 await a discovery about canine semantics that may never come, because there may be nothing of that kind to discover.

Table 9: Selected milestones in cross-species communication decoding. "Type" distinguishes peer-reviewed structure results (R), generative / foundation models (M), interactive grounding protocols (I), and consumer products (P). These advance the toolbox; none yet delivers validated sentence-level translation for any species.

Year	Milestone	Type	Significance
2002	Bow-Lingual [32]	P	Six-emotion bark matcher; Ig Nobel; the consumer archetype
2005	ELTE bark psychophysics [1]	R	Humans classify bark context above chance
2013	DogFACS [6]	R	Objective facial-action coding for dogs
2018	ESP founded; ICASSP bark ML [5]	M/R	Open infrastructure; automated context/affect
2022	CETI program paper [20]	R	Roadmap for whale decoding
2023	AVES; BEANS [24, 27]	M	Self-supervised animal-sound encoder + benchmark
2024	Whale phonetic alphabet [19]	R	Combinatorial coda structure (form, not meaning)
2024	Soundboard studies [15, 16]	R/I	Canine word comprehension + intentional combination
2024	NatureLM-audio [25]	M	Audio–language foundation model for bioacoustics
2025	DolphinGemma + CHAT [21]	M/I	Sequence model + object-grounded interaction
2025	WhAM [22]	M	Generative whale-coda model
2026	ZF-AIM finch interaction [29]	I	Real-time turn-taking with a songbird

11 The Commercial Landscape: Translators That Are Not Translators

A small industry now sells the folkloric dream as a feature. This section catalogs the principal products, reports their claims *as claims*, and audits what they actually do against the ladder. The pattern is uniform: the technology is affect classification (Rung 2); the marketing is dialogue (Rung 5); and the accuracy numbers, as defined, cannot be checked. Table 10 summarizes the landscape.

11.1 A brief history: from Bow-Lingual to PetGPT

The archetype is Takara's Bow-Lingual (2002): a collar microphone and handheld display that matched bark acoustics to six emotions—happiness, sadness, frustration, anger, assertion, desire—and then to one of roughly 200 canned phrases [32]. It sold well, won the 2002 Ig Nobel Peace Prize, and established the template: *classify affect, then decorate it with a sentence.* Two decades later the template is unchanged; only the classifier has been swapped for a deep model and the phrase bank for a language model.

The current flagship is Traini, a Silicon Valley "pet empathy AI" whose PetGPT/PEBI model analyzes sound, images, and video to infer canine emotion and need [31]. Vendor materials report recognition of about 12 emotions across roughly 120 breeds, with a headline accuracy of 81.5%, and describe a corpus on the order of hundreds of thousands of dogs' sounds and expressions labeled with input from behavior specialists. The company has raised venture funding and amassed a large consumer user base. Notably, its founder has publicly conceded that audio alone is insufficient and that body language and facial expression must be fused to reach the claimed accuracy—an implicit admission that the product is multimodal affect inference. The reverse direction (human speech rendered as "dog") is reported to be far weaker, on the order of 40% in independent coverage, and rests on the questionable premise that synthesized barks carry meaning to the dog.

11.2 The wider field

Other products fill out the picture. MeowTalk, from a former Amazon engineer, applies the same idea to cats [33]. The Petpuls collar, a Seoul National University spin-off, classifies emotional states from vocalizations [35]. In China, "Meng Xiao Yi" reports a Qwen-based model trained on millions of pet voiceprints with a claimed 94.6% emotion accuracy [34], and vendors such as PETKIT and various "pet LLM" startups market health-and-emotion monitoring. DogTok and similar apps invert the direction, turning human commands into bark-like sounds for training. The market context is a global pet industry measured in hundreds of billions of dollars, with smart devices the fastest-growing segment—which explains the velocity of the claims, if not their validity.

11.3 The audit: what the accuracy numbers actually measure

Read carefully, every headline number is an *emotion-classification* metric, not a translation metric. "81.5%" means: on some held-out set, the model's emotion label agreed with a human-assigned label 81.5% of the time. That is a real, falsifiable Rung-2 result—*if* the test set, label protocol, and class balance are disclosed, which consumer vendors rarely do. It says nothing about whether the rendered sentence ("I love you forever") corresponds to anything the dog intended, because no such ground truth exists. The sentence is generated *downstream* of the classifier by a language model optimizing for human emotional resonance, not for fidelity to the animal. This is the commercial instance of the semantic gap: the product demonstrates Rung 2 and sells Rung 5.

11.4 Why the gap persists

Three incentives keep the gap open. First, the consumer value proposition is emotional—owners want to feel understood by their pets—so a plausible, affectionate sentence beats a calibrated

probability vector, even though the latter is the honest output. Second, the data flywheel favors engagement over validation: every uploaded clip enriches the classifier, but no mechanism tests whether the "translation" is correct, because the dog cannot complain. Third, the scientific validation loop (playback, blind protocols) is slow and unglamorous, and yields no shareable screenshot. The result is a market optimized for the *appearance* of translation. Section 12 formalizes why this appearance cannot, as presently constructed, be falsified.

Table 10: Commercial "pet translators" audited against the ladder. Accuracies are *vendor-reported claims*, not independently verified; all measure emotion classification (Rung 2), none measures sentence-level translation. "Reverse" = human-to-animal direction.

Product	Modality / method	Claim	Actual rung	Reverse
Bow-Lingual (2002) [32]	Bark acoustics → 6 emotions	∼200 phrases	2	No
Traini / PetGPT [31]	Audio + image + video, PEBI	81.5%, 12 emotions	2	Weak (∼40%)
MeowTalk [33]	Cat vocalization classifier	Emotion labels	2	No
Petpuls [35]	Collar vocalization analysis	Emotion states	2	No
Meng Xiao Yi [34]	Qwen + voiceprints	94.6% (claimed)	2	No
DogTok	Human command → bark	Training aid	n/a	Yes

12 The Evaluation Crisis: Why the Numbers Cannot Be Trusted

If the ladder tells us *what* could be translated, the evaluation problem tells us why we cannot yet know *whether* it has been. This section argues that the field's most quoted quantities—vendor accuracies above 80%, press-release superlatives—are, as presently defined, *unfalsifiable*, and that fixing this is a prerequisite for calling any system a translator.

12.1 The ground-truth problem

Translation, in the human sense, is scored against a reference: a bilingual corpus supplies the intended meaning, and metrics such as BLEU or human adequacy judgments measure fidelity to it. Canine communication has no such reference. There is no canonical "intended message" behind a bark, no bilingual informant who can confirm that "I want to play" is what the dog meant. The dog cannot report its intent, and—per Section 7—there may be no sentence-shaped intent to report. Consequently, a bark-to-sentence system cannot be scored for translational fidelity at all. What *can* be scored is the upstream classifier: does the model's emotion label agree with a human consensus label on a held-out recording? That is a Rung-2 metric, and it is the only honest number in the pipeline. Reporting it as a "translation accuracy" is a category mistake.

12.2 What the vendor numbers actually are

When a product advertises "81.5% accuracy," the defensible reading is: on some internal test set, the predicted emotion class matched a human-assigned class 81.5% of the time. Three problems follow. First, the test set, label protocol, class balance, and inter-annotator agreement are rarely disclosed, so even the Rung-2 claim cannot be independently checked. Second, the label is the *human-perceived* emotion or the *eliciting context*, which is a proxy for the dog's state, not a readout of it; the ELTE tradition is explicit about this [2, 5]. Third, the rendered sentence is produced *after* classification by a language model optimized for human resonance, so its fidelity is never measured at all. The number thus travels down the pipeline as marketing while its meaning stays fixed at the top.

12.3 Unfalsifiability and the absent complaint

A scientific claim must be falsifiable: there must be some observation that could show it false. "This bark means 'I love you' " admits no such observation—the dog will not contradict it, and any subsequent behavior can be retrofitted to the narrative. This is the structural reason the consumer market cannot self-correct: there is no error signal. Compare a weather forecast, which is scored against tomorrow's weather, or a speech-to-text system, scored against a transcript. The pet translator has no tomorrow and no transcript. Until a validation loop supplies an error signal, its outputs are unfalsifiable interpretations, not measurements.

12.4 Clever Hans, anthropomorphism, and the observer effect

Two classic threats compound the problem. The *Clever Hans* effect—the horse that appeared to do arithmetic but was reading its trainer's unconscious cues—is the permanent hazard of any animal-communication experiment: the human observer leaks the answer. Soundboard studies guard against this with blind protocols and unfamiliar button-pressers [15]; consumer apps, used by the devoted owner in an uncontrolled home, do the opposite. *Anthropomorphism* is the second threat: attributing human propositional attitudes ("she says she loves me") to graded affective signals. The critical-review literature on canine emotion perception warns that human observers project their own categories onto the dog [9], and a language model trained on human text will amplify exactly this projection. The product thus risks teaching owners to *ignore* the dog's real body signals in favor of a flattering sentence—a welfare harm we return to in Section 14.

12.5 What a valid evaluation would require

A defensible evaluation program has four ingredients.

(1) **Defined labels and disclosed protocols.** Report Rung-2 metrics on a public, held-out corpus with inter-annotator agreement, class balance, and per-class confusion—the BEANS/BirdSet discipline applied to dogs [27].

(2) **Calibration, not just accuracy.** Output probabilities that match observed frequencies, so a welfare tool can threshold responsibly.

(3) **Closed-loop validation.** Test inferred intent against the dog's subsequent behavior under blind conditions—the playback logic of Section 9, adapted to the home.

(4) **Separate the sentence from the score.** Never report a generative rendering as evidence of fidelity; treat it as a user-interface layer, clearly labeled as interpretation.

None of today's consumer products satisfies these. The research community satisfies (1) and (2) in patches and (3) in the soundboard and cross-species literature. The gap between the two is the evaluation crisis.

12.6 The limitations, consolidated

The evaluation crisis is not unique to pets; it is the animal-communication instance of a broader reckoning in AI about what benchmarks measure and what models merely memorize. The "stochastic parrots" critique warns that scaling text models amplifies form without grounding [150], and the "climbing toward NLU" position paper argues that the field mistakes benchmark gains for understanding [149]. Broad evaluation suites—HELM, BIG-bench, MMLU, TruthfulQA—expose how much headline scores depend on task choice and contamination [153, 154, 155, 156]. Documentation practices such as datasheets and model cards exist precisely to make a system's limits legible [151, 152], and the hallucination/faithfulness literature shows that fluent generation routinely outruns its evidence [157, 158]. Table 11 consolidates the limitations this survey has identified, each with its locus and its consequence.

Table 11: Consolidated limitations of canine "translation," with the locus and consequence of each. This is the survey's audit in one place.

Limitation	Where it bites	Consequence
No ground-truth message	bark-to-sentence scoring	fidelity is unfalsifiable
Human-perceived labels	Rung-2 metrics	a proxy for state, not a readout
Undisclosed protocols	vendor accuracies	claims cannot be checked
Clever Hans / anthropomorphism	uncontrolled home use	the observer leaks the answer
No parallel corpus	supervised training	no translation signal exists
Semantic gap / no recursion	Rung 5	the target may not exist
Fluency $\neq$ fidelity	LLM sentence rendering	hallucinated "translations"
No closed-loop error signal	consumer market	the field cannot self-correct

12.7 A falsifiability criterion for the field

We propose a simple demarcation: *a canine communication system should be called a translator only if its outputs are scored against an independent, animal-grounded error signal.* By this criterion, no current product qualifies; the soundboard paradigm qualifies at the word level; and the rest is affect classification with a narrative interface. This is not pedantry. It is the difference between a measurement and a story, and the field's credibility—and the animals' welfare—depends on keeping them apart.

13 So How Far Are We? An Honest Distance Estimate

We can now answer the title question without evasion. The answer is not a number of years but a map: the distance depends entirely on which rung of the ladder one means, and the

rungs differ not by degree but by kind. Table 12 gives our estimate, with the evidence and the dominant blocker for each.

13.1 Rung 1–2: essentially here, and useful

Signal detection and affect classification are solved in the laboratory and shipping in products. The remaining work is engineering and validation—public benchmarks, calibration, multimodal fusion—not discovery. A calibrated, validated welfare monitor that reports arousal and valence with honest uncertainty is plausibly a **2–4 year** horizon from the research frontier to reliable deployment. This is the genuinely useful payoff: early stress or pain detection, shelter welfare assessment, veterinary triage. It is not "translation," and calling it such would oversell it; but it is real.

13.2 Rung 3: a foothold, a decade of work

Intent and context inference on *natural* signals is emerging. The soundboard literature proves that referential intent exists and is checkable [15, 16], but it does so with human-supplied symbols. Extending this to spontaneous barks, faces, and postures—fusing modalities, modeling goals, validating closed-loop—is a research program, not a product cycle. We estimate **validated natural-signal intent inference at roughly 2030 or later**, contingent on the evaluation infrastructure of Section 12 being built. The blocker is not model capacity; it is labeled, grounded, validated data.

13.3 Rung 4–5: possibly never, and that is informative

Open referential translation and free dialogue presuppose a generative symbolic system in the dog that the biological evidence does not support. Here the honest answer is **no credible timeline**, and the reason is not pessimism about AI but a fact about the subject: there may be no language there to translate. This is the survey's central, deflationary claim. It does not rule out an *engineered* dialogue—structured human-to-dog input via trained symbols, with the dog responding through a soundboard—which is a Rung-3 artifact wearing Rung-5 clothing. But faithful translation of spontaneous canine "speech" into English sentences is, in the strong sense, ill-posed, and more compute will not fix a category error.

13.4 Why "more scale" is not the answer

It is tempting to read the cross-species successes as evidence that scaling foundation models will eventually crack the dog. The reading is mistaken. CETI, DolphinGemma, and the finch models advance *structure discovery and generation*—the front of the pipeline. They do not supply semantics, and their authors say so [19, 21]. The canine bottleneck is at the *back* of the pipeline (grounding, validation) and, above Rung 3, in the subject itself. Scaling the encoder makes Rung 1–2 better; it does not create a grammar that is not there. The distance to Rung 5 is therefore not a function of parameters but of biology.

13.5 The reframe: from translation to interpretation

The productive move is to stop asking "when will we translate dogs?" and start asking "how well can we *interpret* canine state, with calibrated uncertainty and animal-grounded validation?"

That question has answers, has metrics, and has welfare value. It also respects the animal: it treats the dog as a signaler with its own modalities, not as a defective human whose sentences we are waiting to recover. The Rosetta Stone metaphor that titles this survey is, in the end, a warning as much as a promise: the Rosetta Stone worked because the same decree existed in three scripts. For the dog, there may be no decree—only a rich, graded, affective life that we can learn to read, but not to transcribe.

Table 12: Distance estimate by rung. Horizons are judgments from the evidence reviewed, not predictions; "blocker" names the dominant obstacle. Rungs 4–5 are marked ill-posed because the requisite generative symbolic system is not evidenced in natural canine communication.

Rung	Status	Horizon	Dominant blocker
1 Signal	Solved	Now	Engineering only
2 Affect	Shipping	2–4 yr to validated welfare use	Calibration, public benchmarks
3 Intent	Foothold	∼2030+	Grounded, validated data
4 Reference	Unproven	No credible timeline	No productive symbol system
5 Dialogue	Ill-posed	Possibly never	No generative grammar in subject

14 Ethics and Governance: Responsible Interspecies AI

A technology that claims to read an animal's mind carries obligations that a ordinary classifier does not. This section outlines an ethics framework for canine communication AI, organized around welfare, honesty, privacy, and the moral status of the animal as a communication partner.

14.1 Welfare first: do no harm

The most immediate risk is that a flattering "translation" displaces attention to the dog's real signals. An owner told "she says she loves you" may overlook the flattened ears, the whale eye, the lip lick that signal stress; an owner told "my tummy hurts" may either over-react or, worse, dismiss genuine pain when the app stays silent. The critical-review literature on canine emotion perception is explicit that human observers already project their categories onto dogs [9]; a resonant language model amplifies the projection. A welfare-first design therefore (i) surfaces calibrated state estimates with uncertainty, not sentences; (ii) never suppresses or contradicts observable body signals; and (iii) routes high-stakes inferences (pain, aggression, distress) to human or veterinary review. Data collection itself must be non-invasive: passive audio and video, no induced stress to elicit "training" barks.

14.2 Honesty: labeling interpretation as interpretation

The honesty obligation follows directly from the evaluation crisis (Section 12). A system must not present a generative rendering as a measurement. Concretely: separate the calibrated classifier output from the narrative interface; label the latter as interpretation; disclose that accuracy figures refer to emotion classification, not translation; and refrain from reverse-direction claims

(human-to-dog "speech") that have no validation. The Ig Nobel history of Bow-Lingual is a reminder that the market rewards the appearance of understanding; governance should reward the substance.

14.3 Privacy: the home as sensor

Canine monitors live in the home, always-on, capturing audio and video that inevitably include humans. The data flywheel that improves the classifier also accumulates intimate domestic recordings. Privacy-by-design—on-device inference where possible (the DolphinGemma-on-Pixel precedent is instructive [21]), minimal retention, explicit consent for cloud training, and clear deletion—is not optional. The animal cannot consent; the human household must, and the design should make that consent meaningful.

14.4 The animal as partner, not oracle

Finally, there is a conceptual obligation. Treating the dog as an oracle whose "words" we extract instrumentalizes it and flattens its communicative life into a human format. The soundboard paradigm is ethically attractive precisely because it treats the dog as a *partner* in a shared, negotiated symbol system, with the dog's behavior as the check [15, 16]. The cross-species community is beginning to formalize this: CETI's leadership has stressed that any breakthrough in whale communication should be a collective human decision, with responsible-interaction guidelines developed before signals are sent to wild animals. The canine analogue is to validate through the dog's response, to avoid anthropomorphic over-claiming, and to keep the animal's welfare as the terminal criterion.

14.5 Situated in AI ethics and animal sentience

This framework does not stand alone. It inherits the consensus principles of responsible AI: the global landscape of ethics guidelines converges on transparency, justice, non-maleficence, responsibility, and privacy [192], and the gap between principles and practice is well documented [193, 194]. It inherits, too, the political-economy critique that AI systems extract data, labor, and energy from hidden supply chains [195], and the emerging regulatory baseline of risk-tiered obligations such as the EU AI Act [196]. On the animal side, the precautionary principle for sentience argues that credible evidence of affective experience warrants moral consideration even under uncertainty [198], and recent work locates the "edge" of sentience in exactly the graded, affective capacities this survey treats as decodable [197]. The synthesis is a double obligation: be honest about what the model *knows* (AI ethics) and be careful toward what the animal *feels* (sentience ethics). A pet translator that fails either is not a neutral product; it is a welfare and credibility harm.

14.6 A governance checklist

We summarize the framework as a checklist for builders and reviewers:

(1) Report Rung-2 metrics on public, held-out data with calibration; never label them "translation accuracy."

(2) Separate interpretation (narrative) from measurement (classifier); label both clearly.

(3) Validate intent claims closed-loop and blind; otherwise call them hypotheses.

(4) Collect passively; infer on-device where feasible; minimize retention; obtain meaningful household consent.

(5) Route high-stakes outputs to human/veterinary review; never override observable body signals.

(6) Treat the animal as a partner; keep welfare as the terminal criterion.

None of this slows the useful science; all of it protects the animals and the credibility of the field.

15 A Research Agenda: What Would Actually Move the Needle

If the distance estimate of Section 13 is right, then the highest-leverage work is not bigger models but better *infrastructure and protocols*. We propose a concrete agenda, ordered by leverage.

15.1 A public, grounded canine corpus

The field lacks what BEANS and BirdSet provide for general bioacoustics: a standardized, public, multimodal canine corpus with agreed labels [27]. The ideal corpus pairs synchronized audio, video (DogFACS-codeable face and posture), context metadata, and—crucially—*consequent behavior*, so that intent hypotheses can be checked. The soundboard citizen-science network, with thousands of instrumented homes, is the natural collection substrate [15]. Building this corpus with disclosed protocols and inter-annotator agreement would do more for the field than any single model release.

15.2 Calibrated, multimodal affect models

On the modeling side, the priority is fusion and calibration: combine bark, growl, face, and posture into a single state estimate with honest uncertainty, trained and evaluated on the public corpus. Self-supervised encoders (AVES, NatureLM-audio, Perch) supply the backbone [24, 25, 26]; the contribution is the canine-specific grounding and the calibration layer that makes the output safe for welfare use.

15.3 Closed-loop intent validation

The methodological frontier is the validation loop: infer an intent, act on it (or emit a candidate signal), and test the dog's response under blind conditions. The soundboard and CHAT protocols are the templates [16, 21]. A home-adapted, owner-blind protocol that scores intent inference against the dog's subsequent behavior would supply the error signal the field lacks (Section 12) and turn Rung 3 from hypothesis into measurement.

15.4 Structure discovery, with humility

Applying sequence-structure methods to bark corpora is worth doing—but with the explicit prior that canine displays are graded and may yield no clean combinatorial alphabet. The CETI and DolphinGemma toolkits are available [19, 21]; the discipline is to report structure as form, not meaning, and to heed the acoustic-similarity critique [30]. A negative result—no discrete inventory—would itself be valuable, sharpening the semantic-gap claim.

15.5 Engineered dialogue, not extracted dialogue

The most promising path to anything resembling conversation is *engineered*: expand the soundboard paradigm into a richer, negotiated symbol system, with the dog as partner and the behavior as check. This is Rung-3 work that may eventually look like dialogue, without pretending that spontaneous barks contain sentences. It is also the ethically preferable path (Section 14).

15.6 What would falsify our thesis

A good survey states what would change its mind. Our central claim—that Rungs 4–5 are ill-posed for natural canine communication—would be undermined by: (i) a reproducible demonstration of a productive, open referential vocabulary in spontaneous dog signals, validated by playback; or (ii) evidence of recursive, generative combination in natural canine displays (not trained soundboard sequences). Absent such evidence, the ladder stands. We invite the community to try to falsify it; that is how the distance estimate should shrink.

16 Conclusion

We began with a folkloric question—how far are we from translating what a dog says?—and end with a deflationary but, we hope, more useful answer. "Translation" is not one capability but five, and the distance to each is different in kind, not merely in degree. Signal detection and affect classification (Rungs 1–2) are here and shipping; validated affect-and-intent inference (Rung 3) is a research program with a foothold in the soundboard literature and a plausible horizon around 2030; open referential translation and free dialogue (Rungs 4–5) presuppose a generative symbolic system that the biological evidence does not support, and may be permanently ill-posed. The canine Rosetta Stone is, in the end, a metaphor with a catch: the original worked because one decree existed in three scripts, and for the dog there may be no decree to transcribe—only a rich, graded, affective life we can learn to read but not to render as sentences.

Three findings anchor this verdict. First, the biology: dogs signal affect, arousal, identity, motivation, and—with human scaffolding—referential intent, but they do not possess an open-ended generative grammar [1, 4, 6, 16]. Second, the technology: the decoding pipeline's front (encoding, structure discovery) is maturing fast on the back of the cross-species foundation-model boom, while its back (grounding, validation) is where canine-specific work is needed and where commercial effort is absent [24, 25, 19, 21]. Third, the evaluation: the field's headline accuracies are unfalsifiable as translation metrics, because no ground-truth intended message exists to score against; they are, at best, Rung-2 emotion-classification numbers [5]. The commercial market demonstrates Rung 2 and sells Rung 5, and the gap between those is the subject of this survey.

The way forward is not more scale but better infrastructure and protocols: a public grounded canine corpus, calibrated multimodal affect models, closed-loop blind validation, and an ethics framework that treats the animal as a partner and keeps welfare as the terminal criterion. The useful, achievable goal is *interpretation with calibrated uncertainty*, not transcription of a language that may not exist. That goal is within reach this decade, and it is worth reaching.

16.1 Limitations of this survey

In the spirit of evidence-bounded research, we state the limitations plainly. (1) This is a narrative survey, not a PRISMA systematic review; sources were gathered through structured web and literature search in November 2026, and coverage is broad rather than exhaustive. (2) Several fast-moving items (WhAM, ZF-AIM, vendor metrics) rest on project reports and journalism rather than peer review, and are labeled as such; commercial accuracies are reported *as claims* and were not independently verified. (3) Some citations were confirmed to the level of venue, year, and DOI via secondary indexes rather than full-text retrieval; a production version should verify each against the primary record. (4) The distance estimates are expert judgments from the reviewed evidence, not forecasts, and Section 15 states what would falsify them. These limitations do not weaken the central claim—they bound it, which is the point.

Acknowledgments and tools

This survey was prepared with the assistance of open research-agent tooling, including the Scientific Agent Skills literature-review and autoresearch workflows [38], which contributed the search-documentation and citation-verification discipline used here. The figures are original TikZ constructions released with the repository for reuse and correction.

Data and code availability

The LaTeX source, both hero figures, the bibliography, and the project page are openly available in the `woof-to-word` repository: `https://github.com/mcxiaoxiao/woof2word`. Corrections and falsification attempts are welcome.

References

References

[1] P. Pongrácz, C. Molnár, A. Miklósi, and V. Csányi, "Human listeners are able to classify dog (*Canis familiaris*) barks recorded in different situations," *Journal of Comparative Psychology*, vol. 119, no. 2, pp. 136–144, 2005.

[2] P. Pongrácz, C. Molnár, and A. Miklósi, "Acoustic parameters of dog barks carry emotional information for humans," *Applied Animal Behaviour Science*, vol. 100, no. 3–4, pp. 228–240, 2006.

[3] C. Molnár, P. Pongrácz, A. Dóka, and A. Miklósi, "Can humans discriminate between dogs on the basis of the acoustic parameters of barks?" *Behavioural Processes*, vol. 73, no. 1, pp. 76–83, 2006.

[4] T. Faragó, N. Takács, A. Miklósi, and P. Pongrácz, "Dog growls express various contextual and affective content for human listeners," *Royal Society Open Science*, vol. 4, p. 170134, 2017.

[5] "What is my dog trying to tell me? The automatic recognition of the context and perceived emotion of dog barks," in *Proc. IEEE ICASSP*, 2018. DOI: 10.1109/ICASSP.2018.8461757.

[6] B. M. Waller, K. Peirce, C. C. Caeiro, et al., "DogFACS: The dog facial action coding system," 2013. Available: `https://www.animalfacs.com`.

[7] J. Kaminski, J. Waller, R. Diogo, A. Hartstone-Rose, and A. M. Burrows, "Human attention affects facial expressions in domestic dogs," *Scientific Reports*, vol. 7, p. 13066, 2017.

[8] "Domestication constrains the ability of dogs to convey emotions via facial expressions in comparison to their wolf ancestors," *Scientific Reports*, 2024. DOI: 10.1038/s41598-024-61110-6.

[9] "Explainable automated recognition of emotional states from canine facial expressions," *Scientific Reports*, 2022. DOI: 10.1038/s41598-022-27079-w.

[10] "Pet dog facial expression recognition based on convolutional neural network and improved whale optimization algorithm," *Scientific Reports*, 2023. DOI: 10.1038/s41598-023-30442-0.

[11] "Automated analysis of emotional expressions in dogs," *Scientific Reports*, 2025. DOI: 10.1038/s41598-025-15741-y.

[12] "Dog facial landmarks detection and its applications for facial analysis," *Scientific Reports*, 2025. DOI: 10.1038/s41598-025-07040-3.

[13] J. Kaminski, J. Call, and J. Fischer, "Word learning in a domestic dog: Evidence for "fast mapping"," *Science*, vol. 304, no. 5677, pp. 1682–1683, 2004.

[14] J. W. Pilley and A. K. Reid, "Border collie comprehends object names as verbal referents," *Behavioural Processes*, vol. 86, no. 2, pp. 184–195, 2011.

[15] A. P. M. Bastos, A. Evenson, P. M. Wood, et al., "How do soundboard-trained dogs respond to human button presses? An investigation into word comprehension," *PLoS ONE*, vol. 19, p. e0307189, 2024. DOI: 10.1371/journal.pone.0307189.

[16] A. P. M. Bastos, Z. N. Houghton, L. Naranjo, and F. Rossano, "Soundboard-trained dogs produce non-accidental, non-random and non-imitative two-button combinations," *Scientific Reports*, vol. 14, p. 28771, 2024. DOI: 10.1038/s41598-024-79517-6.

[17] "Let's talk about "talking" dogs! Reviewing the science of soundboard use," *Animal Cognition*, 2025. PMC12367964.

[18] A. P. M. Bastos and F. Rossano, "Soundboard-using pets?" *Interaction Studies*, vol. 24, no. 2, pp. 311–334, 2023. DOI: 10.1075/is.22050.pin.

[19] P. Sharma, S. Gero, R. Payne, D. F. Gruber, D. Rus, A. Torralba, and J. Andreas, "Contextual and combinatorial structure in sperm whale vocalisations," *Nature Communications*, vol. 15, p. 3617, 2024. DOI: 10.1038/s41467-024-47221-8.

[20] J. Andreas, G. Beguš, M. M. Bronstein, et al., "Toward understanding the communication in sperm whales," *iScience*, vol. 25, p. 104393, 2022.

[21] Google DeepMind, Georgia Tech, and Wild Dolphin Project, "DolphinGemma: How Google AI is helping decode dolphin communication," Project report, 14 April 2025.

[22] Project CETI, "WhAM: Whale Acoustics Model," Technical report, 2025.

[23] "Towards a translative model of sperm whale vocalization," *arXiv:2512.02206*, 2025.

[24] M. Hagiwara, "AVES: Animal vocalization encoder based on self-supervision," in *Proc. IEEE ICASSP*, 2023.

[25] D. Robinson, M. Miron, M. Hagiwara, and O. Pietquin, "NatureLM-audio: An audio–language foundation model for bioacoustics," *arXiv:2411.07186*, 2024.

[26] B. Ghani, T. Denton, S. Kahl, and H. Klinck, "Global birdsong embeddings enable superior transfer learning for bioacoustic classification," *Scientific Reports*, vol. 13, p. 22876, 2023.

[27] M. Hagiwara et al., "BEANS: The benchmark of animal sounds," in *Proc. IEEE ICASSP*, 2023.

[28] "OpenBEATs: A fully open-source general-purpose audio encoder," *arXiv:2507.14129*, 2025.

[29] Earth Species Project and McGill University, "ZF-AIM: An interactive zebra-finch vocalization model," Technical report, 2026 (not yet peer reviewed).

[30] Y. Yovel, Commentary on acoustic-similarity pitfalls in animal-language models, reported in *Nature* news coverage, 2025–2026.

[31] Traini (PetGPT / PEBI model), Commercial pet-empathy AI, 2024–2026. Vendor-reported metrics; see Section 11.

[32] Takara Co., "Bow-Lingual" dog-bark translator, 2002. Ig Nobel Peace Prize, 2002.

[33] J. Sanchez (Akvelon), "MeowTalk" cat-vocalization app, 2020.

[34] "Meng Xiao Yi" pet translator (Alibaba Qwen-based; vendor-claimed 94.6%), 2026. Reported via Chinese technology press.

[35] Petpuls AI collar (Seoul National University spin-off), 2021.

[36] A. Quaranta, M. Siniscalchi, and G. Vallortigara, "Asymmetric tail-wagging responses by dogs to different emotive stimuli," *Current Biology*, vol. 17, no. 6, pp. R199–R201, 2007.

[37] A. Andics, A. Gácsi, T. Faragó, A. Kis, and A. Miklósi, "Voice-sensitive regions in the dog and human brain are revealed by comparative fMRI," *Current Biology*, vol. 24, no. 5, pp. 574–578, 2014.

[38] T. Kassis, V. Agarwal, Y. He, D. Patel, and A. M. Brueckner, "Scientific Agent Skills: A library of procedural knowledge for research agents," *arXiv:2609.00065*, 2026. DOI: 10.48550/arXiv.2609.00065.

[39] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to sequence learning with neural networks," in *NeurIPS*, 2014.

[40] K. Cho et al., "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *EMNLP*, 2014.

[41] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," in *ICLR*, 2015.

[42] M.-T. Luong, H. Pham, and C. D. Manning, "Effective approaches to attention-based neural machine translation," in *EMNLP*, 2015.

[43] J. Gehring et al., "Convolutional sequence to sequence learning," in *ICML*, 2017.

[44] A. Vaswani et al., "Attention is all you need," in *NeurIPS*, 2017. arXiv:1706.03762.

[45] Y. Wu et al., "Google's neural machine translation system: Bridging the gap between human and machine translation," arXiv:1609.08144, 2016.

[46] M. Johnson et al., "Google's multilingual neural machine translation system: Enabling zero-shot translation," *TACL*, vol. 5, pp. 339–351, 2017.

[47] O. Firat, K. Cho, and Y. Bengio, "Multi-way, multilingual neural machine translation with a shared attention mechanism," in *NAACL*, 2016.

[48] M. Ott et al., "Scaling neural machine translation," in *WMT*, 2018.

[49] S. Edunov et al., "Understanding back-translation at scale," in *EMNLP*, 2018.

[50] R. Sennrich, B. Haddow, and A. Birch, "Neural machine translation of rare words with subword units," in *ACL*, 2016.

[51] R. Sennrich, B. Haddow, and A. Birch, "Improving neural machine translation models with monolingual data," in *ACL*, 2016.

[52] P. Koehn et al., "Moses: Open source toolkit for statistical machine translation," in *ACL (demo)*, 2005.

[53] P. Koehn, "Europarl: A parallel corpus for statistical machine translation," in *MT Summit*, 2005.

[54] J. Tiedemann, "Parallel data, tools and interfaces in OPUS," in *LREC*, 2012.

[55] K. Papineni et al., "BLEU: A method for automatic evaluation of machine translation," in *ACL*, 2002.

[56] M. Denkowski and A. Lavie, "Meteor universal: Language specific translation evaluation for any target language," in *EACL*, 2014.

[57] R. Rei, C. Stewart, and A. Farinha, "Unbabel's participation in the WMT20 metrics shared task (COMET)," in *WMT*, 2020.

[58] M. Freitag et al., "Experts, errors, and context: A large-scale study of human evaluation for machine translation," *TACL*, vol. 9, pp. 146–164, 2021.

[59] O. Bojar et al., "Findings of the 2018 conference on machine translation (WMT18)," in *WMT*, 2018.

[60] L. Barrault et al., "Findings of the 2019 conference on machine translation (WMT19)," in *WMT*, 2019.

[61] T. Kocmi et al., "Findings of the 2022 conference on machine translation (WMT22)," in *WMT*, 2022.

[62] M. R. Costa-jussà et al., "No language left behind: Scaling human-centered machine translation," arXiv:2207.04672, 2022.

[63] NLLB Team, "Scaling neural machine translation to 200 languages," *Nature*, vol. 630, pp. 841–846, 2024.

[64] A. Fan et al., "Beyond English-centric multilingual machine translation," *JMLR*, vol. 22, pp. 1–48, 2021.

[65] M. Artetxe et al., "Unsupervised neural machine translation," in *ICLR*, 2018.

[66] G. Lample et al., "Phrase-based & neural unsupervised machine translation," in *EMNLP*, 2018.

[67] H. Schwenk et al., "WikiMatrix: Mining 135M parallel sentences in 1620 language pairs from Wikipedia," in *ACL*, 2021.

[68] A. El-Kishky et al., "CCAligned: A massive collection of cross-lingual web-document pairs," in *LREC*, 2020.

[69] A. Bérard et al., "Listen and translate: A proof of concept for end-to-end speech-to-text translation," in *NeurIPS Workshop*, 2016.

[70] R. J. Weiss et al., "Sequence-to-sequence models can directly translate foreign speech," in *Interspeech*, 2017.

[71] Y. Jia et al., "Direct speech-to-speech translation with a sequence-to-sequence model," in *Interspeech*, 2019.

[72] Seamless Communication et al., "SeamlessM4T: Massively multilingual & multimodal machine translation," arXiv:2308.11596, 2023.

[73] Seamless Communication et al., "Seamless: Multilingual expressive and streaming speech translation," arXiv:2312.05187, 2023.

[74] Y. Liu et al., "Multilingual denoising pre-training for neural machine translation (mBART)," *TACL*, vol. 8, pp. 726–742, 2020.

[75] T. Mikolov et al., "Efficient estimation of word representations in vector space," arXiv:1301.3781, 2013.

[76] J. Pennington, R. Socher, and C. D. Manning, "GloVe: Global vectors for word representation," in *EMNLP*, 2014.

[77] M. E. Peters et al., "Deep contextualized word representations (ELMo)," in *NAACL*, 2018.

[78] J. Howard and S. Ruder, "Universal language model fine-tuning for text classification (ULMFiT)," in *ACL*, 2018.

[79] J. Devlin et al., "BERT: Pre-training of deep bidirectional transformers for language understanding," in *NAACL*, 2019. arXiv:1810.04805.

[80] A. Radford et al., "Improving language understanding by generative pre-training," OpenAI, Tech. Rep., 2018.

[81] A. Radford et al., "Language models are unsupervised multitask learners," OpenAI, Tech. Rep., 2019.

[82] T. Brown et al., "Language models are few-shot learners," in *NeurIPS*, 2020. arXiv:2005.14165.

[83] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer (T5)," *JMLR*, vol. 21, pp. 1–67, 2020.

[84] M. Lewis et al., "BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension," in *ACL*, 2020.

[85] Y. Liu et al., "RoBERTa: A robustly optimized BERT pretraining approach," arXiv:1907.11692, 2019.

[86] Z. Yang et al., "XLNet: Generalized autoregressive pretraining for language understanding," in *NeurIPS*, 2019.

[87] K. Clark et al., "ELECTRA: Pre-training text encoders as discriminators rather than generators," in *ICLR*, 2020.

[88] P. He et al., "DeBERTa: Decoding-enhanced BERT with disentangled attention," in *ICLR*, 2021.

[89] A. Conneau et al., "Unsupervised cross-lingual representation learning at scale (XLM-R)," in *ACL*, 2020.

[90] T. Pires, E. Schlinger, and D. Garrette, "How multilingual is multilingual BERT?" in *ACL*, 2019.

[91] A. Conneau and G. Lample, "Cross-lingual language model pretraining (XLM)," in *NeurIPS*, 2019.

[92] H. Touvron et al., "LLaMA: Open and efficient foundation language models," arXiv:2302.13971, 2023.

[93] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," arXiv:2307.09288, 2023.

[94] OpenAI, "GPT-4 technical report," arXiv:2303.08774, 2023.

[95] Gemini Team, "Gemini: A family of highly capable multimodal models," arXiv:2312.11805, 2024.

[96] Anthropic, "The Claude 3 model family: Opus, Sonnet, Haiku," Tech. Rep., 2024.

[97] Gemma Team, "Gemma: Open models based on Gemini research and technology," arXiv:2403.08295, 2024.

[98] L. Ouyang et al., "Training language models to follow instructions with human feedback (InstructGPT)," in *NeurIPS*, 2022.

[99] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," in *NeurIPS*, 2022.

[100] R. Bommasani et al., "On the opportunities and risks of foundation models," arXiv:2108.07258, 2021.

[101] J. Kaplan et al., "Scaling laws for neural language models," arXiv:2001.08361, 2020.

[102] J. Hoffmann et al., "Training compute-optimal large language models (Chinchilla)," in *NeurIPS*, 2022.

[103] A. Radford et al., "Learning transferable visual models from natural language supervision (CLIP)," in *ICML*, 2021.

[104] J.-B. Alayrac et al., "Flamingo: A visual language model for few-shot learning," in *NeurIPS*, 2022.

[105] H. Liu et al., "Visual instruction tuning (LLaVA)," in *NeurIPS*, 2023.

[106] J. Li et al., "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *ICML*, 2023.

[107] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," arXiv:1807.03748, 2018.

[108] S. Schneider et al., "wav2vec: Unsupervised pre-training for speech recognition," in *Interspeech*, 2019.

[109] A. Baevski et al., "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *NeurIPS*, 2020. arXiv:2006.11477.

[110] W.-N. Hsu et al., "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM TASLP*, vol. 29, pp. 3451–3460, 2021.

[111] S. Chen et al., "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE JSTSP*, vol. 16, no. 6, pp. 1505–1518, 2022.

[112] A. Baevski et al., "data2vec: A general framework for self-supervised learning in speech, vision and language," in *ICML*, 2022.

[113] A. Baevski et al., "Efficient self-supervised learning with contextualized target representations (data2vec 2.0)," in *ICML*, 2023.

[114] A. Radford et al., "Robust speech recognition via large-scale weak supervision (Whisper)," in *ICML*, 2023. arXiv:2212.04356.

[115] A. Graves et al., "Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks," in *ICML*, 2006.

[116] A. Graves, A. Mohamed, and G. Hinton, "Speech recognition with deep recurrent neural networks," in *ICASSP*, 2013.

[117] D. Amodei et al., "Deep Speech 2: End-to-end speech recognition in English and Mandarin," in *ICML*, 2016.

[118] W. Chan et al., "Listen, attend and spell: A neural network for large vocabulary conversational speech recognition," in *ICASSP*, 2016.

[119] A. Gulati et al., "Conformer: Convolution-augmented transformer for speech recognition," in *Interspeech*, 2020.

[120] V. Panayotov et al., "Librispeech: An ASR corpus based on public domain audio books," in *ICASSP*, 2015.

[121] R. Ardila et al., "Common Voice: A massively-multilingual speech corpus," in *LREC*, 2020.

[122] V. Pratap et al., "MLS: A large-scale multilingual dataset for speech research," in *Interspeech*, 2020.

[123] J. S. Garofolo et al., "TIMIT acoustic-phonetic continuous speech corpus," Linguistic Data Consortium, 1993.

[124] J. J. Godfrey, E. C. Holliman, and J. McDaniel, "SWITCHBOARD: Telephone speech corpus for research and development," in *ICASSP*, 1992.

[125] S.-w. Yang et al., "SUPERB: Speech processing universal performance benchmark," in *Interspeech*, 2021.

[126] A. Baevski et al., "Unsupervised speech recognition," in *NeurIPS*, 2021.

[127] A. Babu et al., "XLS-R: Self-supervised cross-lingual speech representation learning at scale," in *Interspeech*, 2022.

[128] V. Pratap et al., "Scaling speech technology to 1,000+ languages (MMS)," in *ACL*, 2023.

[129] Y. Zhang et al., "Google USM: Scaling automatic speech recognition beyond 100 languages," arXiv:2303.01037, 2023.

[130] C. Wang et al., "CoVoST: A diverse multilingual speech-to-text translation corpus," in *LREC*, 2020.

[131] C. Wang et al., "CoVoST 2 and massively multilingual speech-to-text translation," in *Interspeech*, 2021.

[132] A. van den Oord et al., "WaveNet: A generative model for raw audio," arXiv:1609.03499, 2016.

[133] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, "Neural discrete representation learning (VQ-VAE)," in *NeurIPS*, 2017.

[134] P. Dhariwal et al., "Jukebox: A generative model for music," arXiv:2005.00341, 2020.

[135] N. Zeghidour et al., "SoundStream: An end-to-end neural audio codec," *IEEE/ACM TASLP*, vol. 30, pp. 495–507, 2021.

[136] A. Défossez et al., "High fidelity neural audio compression (EnCodec)," *TMLR*, 2023. arXiv:2210.13438.

[137] Z. Borsos et al., "AudioLM: A language modeling approach to audio generation," *IEEE/ACM TASLP*, vol. 31, pp. 252–264, 2023.

[138] A. Agostinelli et al., "MusicLM: Generating music from text," arXiv:2301.11325, 2023.

[139] J. Copet et al., "Simple and controllable music generation (MusicGen)," in *NeurIPS*, 2023.

[140] F. Kreuk et al., "AudioGen: Textually guided audio generation," in *ICLR*, 2023.

[141] C. Wang et al., "Neural codec language models are zero-shot text to speech synthesizers (VALL-E)," arXiv:2301.02111, 2023.

[142] D. Yang et al., "UniAudio: An audio foundation model toward universal audio generation," in *ICML*, 2024.

[143] P. K. Rubenstein et al., "AudioPaLM: A large language model that can speak and listen," arXiv:2306.12925, 2023.

[144] C. Tang et al., "SALMONN: Towards generic hearing abilities for large language models," in *ICLR*, 2024.

[145] Y. Chu et al., "Qwen-Audio: Advancing universal audio understanding via unified large-scale audio-language models," arXiv:2311.07919, 2023.

[146] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE TKDE*, vol. 22, no. 10, pp. 1345–1359, 2010.

[147] F. Zhuang et al., "A comprehensive survey on transfer learning," *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, 2021.

[148] J. Yosinski et al., "How transferable are features in deep neural networks?" in *NeurIPS*, 2014.

[149] E. M. Bender and A. Koller, "Climbing towards NLU: On meaning, form, and understanding in the age of data," in *ACL*, 2020.

[150] E. M. Bender et al., "On the dangers of stochastic parrots: Can language models be too big?" in *FAccT*, 2021.

[151] T. Gebru et al., "Datasheets for datasets," *CACM*, vol. 64, no. 12, pp. 86–92, 2021.

[152] M. Mitchell et al., "Model cards for model reporting," in *FAT**, 2019.

[153] P. Liang et al., "Holistic evaluation of language models (HELM)," *TMLR*, 2023.

[154] A. Srivastava et al., "Beyond the imitation game: Quantifying and extrapolating the capabilities of language models (BIG-bench)," *TMLR*, 2023.

[155] D. Hendrycks et al., "Measuring massive multitask language understanding (MMLU)," in *ICLR*, 2021.

[156] S. Lin, J. Hilton, and O. Evans, "TruthfulQA: Measuring how models mimic human falsehoods," in *ACL*, 2022.

[157] Z. Ji et al., "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, pp. 1–38, 2023.

[158] J. Maynez et al., "On faithfulness and factuality in abstractive summarization," in *ACL*, 2020.

[159] S. Harnad, "The symbol grounding problem," *Physica D*, vol. 42, pp. 335–346, 1990.

[160] Y. Bisk et al., "Experience grounds language," in *EMNLP*, 2020.

[161] D. Hupkes et al., "Compositionality decomposed: How do neural networks generalise?" *JAIR*, vol. 67, pp. 757–795, 2020.

[162] B. M. Lake and M. Baroni, "Human-like systematic generalization through a meta-learning neural network," *Nature*, vol. 623, pp. 115–121, 2023.

[163] B. M. Lake and M. Baroni, "Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks," in *ICML*, 2018.

[164] A. Conneau et al., "What you can cram into a single vector: Probing sentence embeddings for linguistic properties," in *ACL*, 2018.

[165] A. Rogers, O. Kovaleva, and A. Rumshisky, "A primer on BERTology: What we know about how BERT works," *TACL*, vol. 8, pp. 842–866, 2020.

[166] S. Kahl et al., "BirdNET: A deep learning solution for avian diversity monitoring," *Ecological Informatics*, vol. 61, p. 101236, 2021.

[167] D. Stowell, "Computational bioacoustics with deep learning: A review and roadmap," *PeerJ*, vol. 10, p. e13152, 2022.

[168] D. Stowell and M. D. Plumbley, "Automatic large-scale classification of bird sounds is strongly improved by unsupervised feature learning," *PeerJ*, vol. 2, p. e488, 2014.

[169] V. M. Janik and P. J. B. Slater, "Context-specific use suggests that bottlenose dolphin signature whistles are cohesion calls," *Animal Behaviour*, vol. 56, pp. 829–838, 1998.

[170] S. L. King and V. M. Janik, "Bottlenose dolphins can use learned vocal labels to address each other," *PNAS*, vol. 110, pp. 13216–13221, 2013.

[171] R. M. Seyfarth, D. L. Cheney, and P. Marler, "Monkey responses to three different alarm calls: Evidence of predator classification and semantic communication," *Science*, vol. 210, pp. 801–803, 1980.

[172] D. L. Cheney and R. M. Seyfarth, *How Monkeys See the World*. University of Chicago Press, 1990.

[173] K. von Frisch, *The Dance Language and Orientation of Bees*. Harvard University Press, 1967.

[174] C. Crockford et al., "Wild chimpanzees inform ignorant group members of danger," *Current Biology*, vol. 22, pp. 142–146, 2012.

[175] T. N. Suzuki, D. Wheatcroft, and M. Griesser, "Experimental evidence for compositional syntax in bird calls," *Nature Communications*, vol. 7, p. 10986, 2016.

[176] P. Schlenker et al., "Formal monkey linguistics," *Theoretical Linguistics*, vol. 42, pp. 1–90, 2016.

[177] C. N. Slobodchikoff, A. Paseka, and J. L. Verdolin, "Prairie dog alarm calls encode labels about predator colors," *Animal Cognition*, vol. 12, pp. 435–439, 2009.

[178] C. S. Evans, L. Evans, and P. Marler, "On the meaning of alarm calls: Functional reference in an avian vocal system," *Animal Behaviour*, vol. 46, pp. 23–38, 1993.

[179] M. D. Hauser, *The Evolution of Communication*. MIT Press, 1996.

[180] M. Tomasello, *Origins of Human Communication*. MIT Press, 2008.

[181] S. Kirby, H. Cornish, and K. Smith, "Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language," *PNAS*, vol. 105, pp. 10681–10686, 2008.

[182] K. Smith, S. Kirby, and H. Brighton, "Iterated learning: A framework for the emergence of language," *Artificial Life*, vol. 9, pp. 371–386, 2003.

[183] M. D. Hauser, N. Chomsky, and W. T. Fitch, "The faculty of language: What is it, who has it, and how did it evolve?" *Science*, vol. 298, pp. 1569–1579, 2002.

[184] S. Pinker, *The Language Instinct*. William Morrow, 1994.

[185] N. Chomsky, *Syntactic Structures*. Mouton, 1957.

[186] G. K. Zipf, *Human Behavior and the Principle of Least Effort*. Addison-Wesley, 1949.

[187] B. Hare et al., "The domestication of social cognition in dogs," *Science*, vol. 298, pp. 1634–1636, 2002.

[188] A. Miklósi et al., "A simple reason for a big difference: Wolves do not look back at humans, but dogs do," *Current Biology*, vol. 13, pp. 763–766, 2003.

[189] K. Soproni et al., "Comprehension of human communicative signs in pet dogs," *Journal of Comparative Psychology*, vol. 115, pp. 122–126, 2001.

[190] G. S. Berns et al., "Functional MRI in awake unrestrained dogs," *PLoS ONE*, vol. 7, p. e38027, 2012.

[191] L. A. F. Frantz et al., "Genomic and archaeological evidence suggest a dual origin of domestic dogs," *Science*, vol. 352, pp. 1228–1231, 2016.

[192] A. Jobin, M. Ienca, and E. Vayena, "The global landscape of AI ethics guidelines," *Nature Machine Intelligence*, vol. 1, pp. 389–399, 2019.

[193] B. Mittelstadt, "Principles alone cannot guarantee ethical AI," *Nature Machine Intelligence*, vol. 1, pp. 501–507, 2019.

[194] L. Floridi et al., "AI4People—An ethical framework for a good AI society," *Minds and Machines*, vol. 28, pp. 689–707, 2018.

[195] K. Crawford, *Atlas of AI*. Yale University Press, 2021.

[196] European Parliament, "Artificial Intelligence Act," Regulation (EU) 2024/1689, 2024.

[197] J. Birch, *The Edge of Sentience*. Oxford University Press, 2024.

[198] J. Birch, "Animal sentience and the precautionary principle," *Animal Sentience*, vol. 2, no. 16, 2017.

[199] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *ICLR*, 2015.

[200] N. Srivastava et al., "Dropout: A simple way to prevent neural networks from overfitting," *JMLR*, vol. 15, pp. 1929–1958, 2014.

[201] K. He et al., "Deep residual learning for image recognition," in *CVPR*, 2016.

[202] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale (ViT)," in *ICLR*, 2021.

[203] I. Goodfellow et al., "Generative adversarial nets," in *NeurIPS*, 2014.

[204] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *NeurIPS*, 2020.

[205] R. Rombach et al., "High-resolution image synthesis with latent diffusion models," in *CVPR*, 2022.

[206] J. Schulman et al., "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[207] V. Mnih et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, pp. 529–533, 2015.

[208] D. Silver et al., "Mastering the game of Go with deep neural networks and tree search," *Nature*, vol. 529, pp. 484–489, 2016.

[209] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015.

[210] Y. Bengio, A. Courville, and P. Vincent, "Representation learning: A review and new perspectives," *IEEE TPAMI*, vol. 35, pp. 1798–1828, 2013.

[211] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[212] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, pp. 533–536, 1986.

[213] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, pp. 1735–1780, 1997.

[214] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *NeurIPS*, 2012.

[215] J. Deng et al., "ImageNet: A large-scale hierarchical image database," in *CVPR*, 2009.

[216] O. Russakovsky et al., "ImageNet large scale visual recognition challenge," *IJCV*, vol. 115, pp. 211–252, 2015.

[217] Dogster, "How many dogs are there? US & worldwide statistics," 2026. [Online]. Available: `https://www.dogster.com/statistics/how-many-dogs-are-there-statistics/`

[218] "Consumer acceptance of sustainable dog diets: A survey," 2025. PMC12560912 (reports a 2024 companion-dog population of ≈528 million).

[219] FEDIAF, "Facts & Figures 2025: European pet population," European Pet Food Industry Federation, 2025.

[220] AVMA, "2025 Pet Ownership and Demographics Sourcebook," American Veterinary Medical Association, 2025.

[221] Budget Direct, "Australian Dog and Cat Care Survey 2024," 2024.

[222] World Animal Foundation, "Pet ownership statistics 2026," 2026.